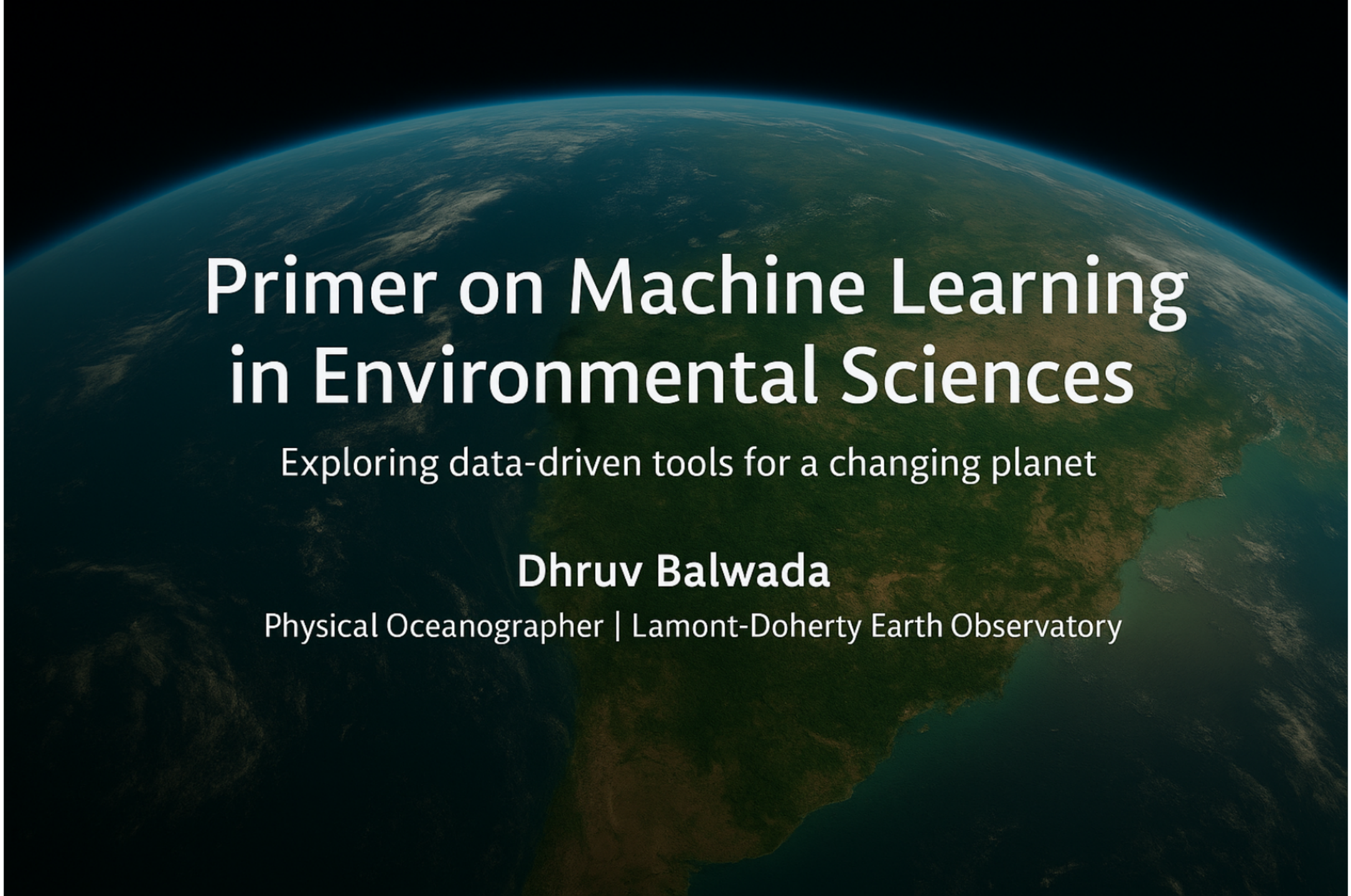
Introduction to Climate Modeling

Week 10: AI and climate modeling

Dhruv Balwada · Columbia Climate School · Fall 2026

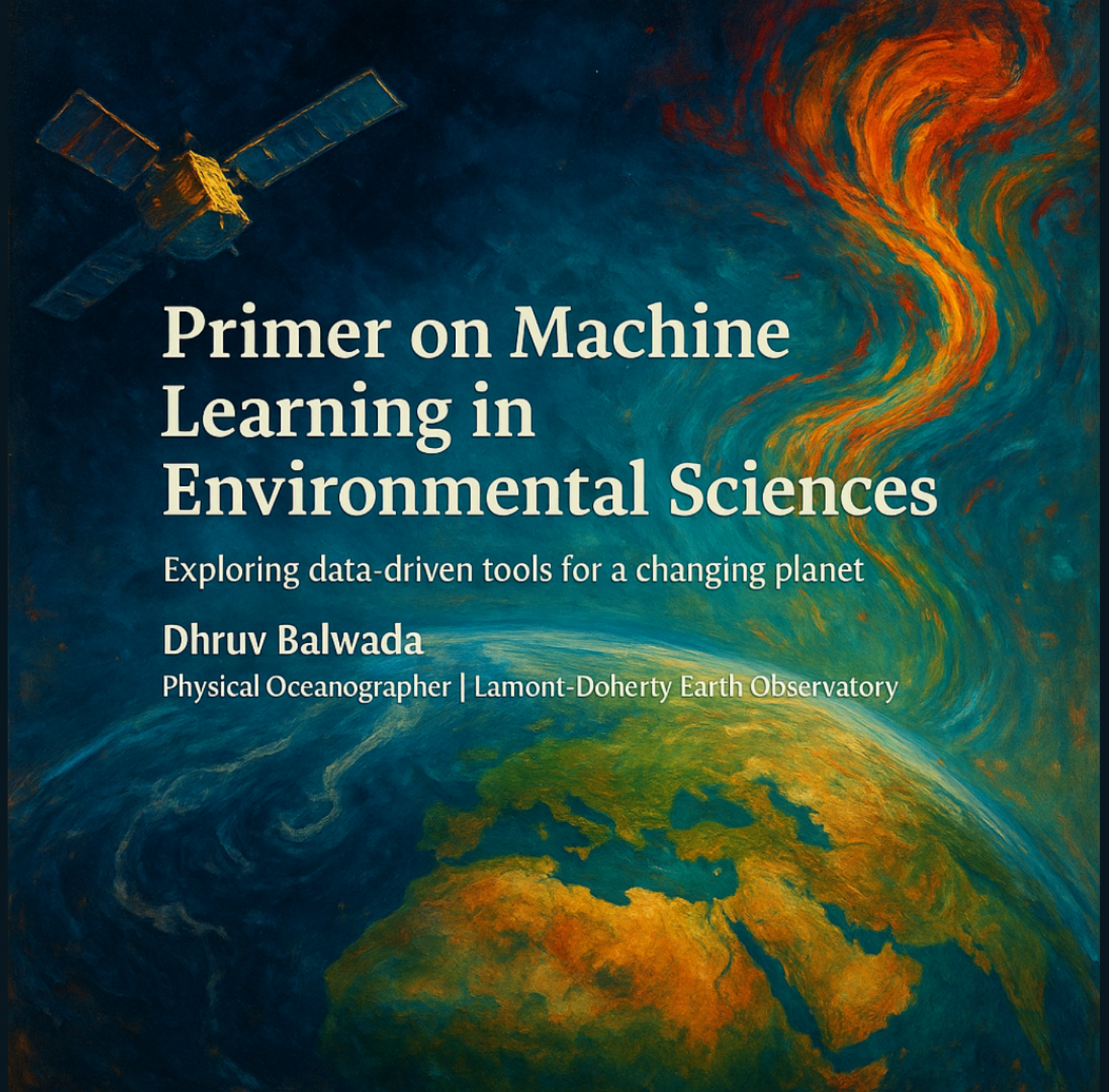


Primer on Machine Learning in Environmental Sciences

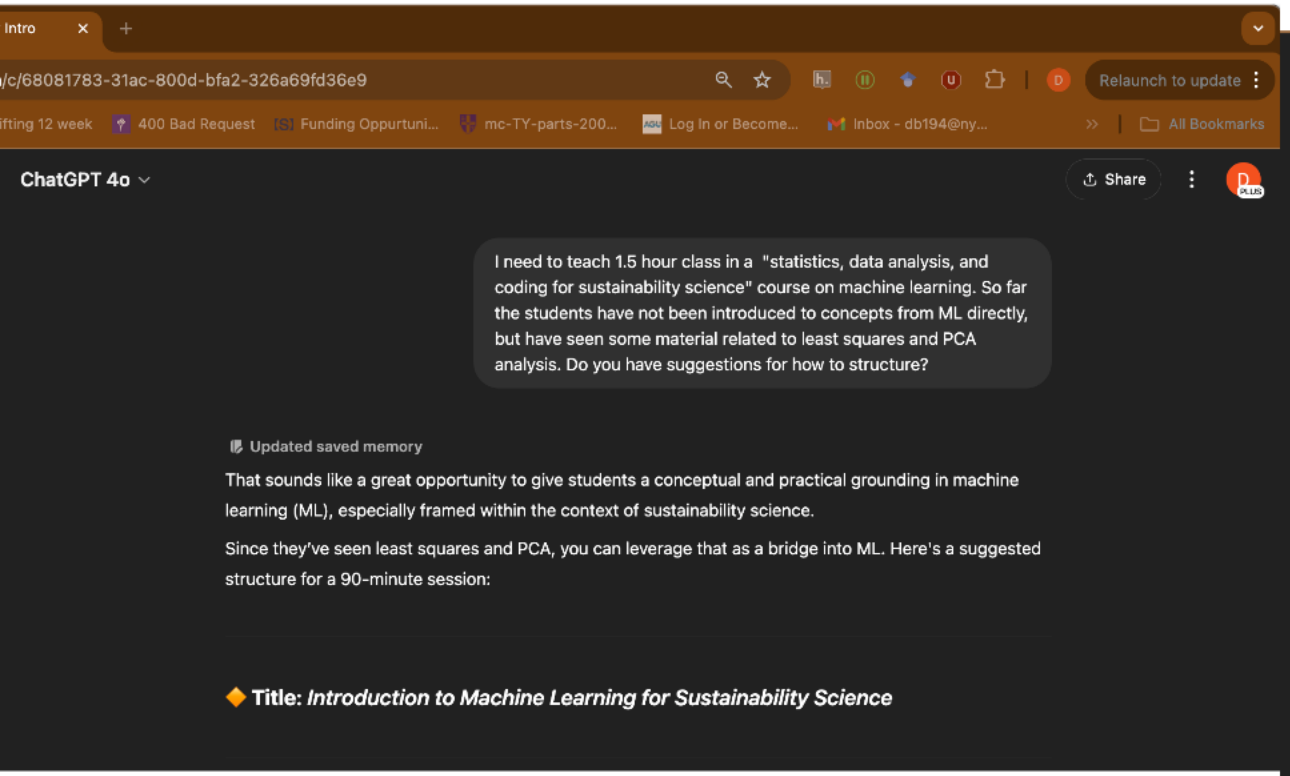
Exploring data-driven tools for a changing planet

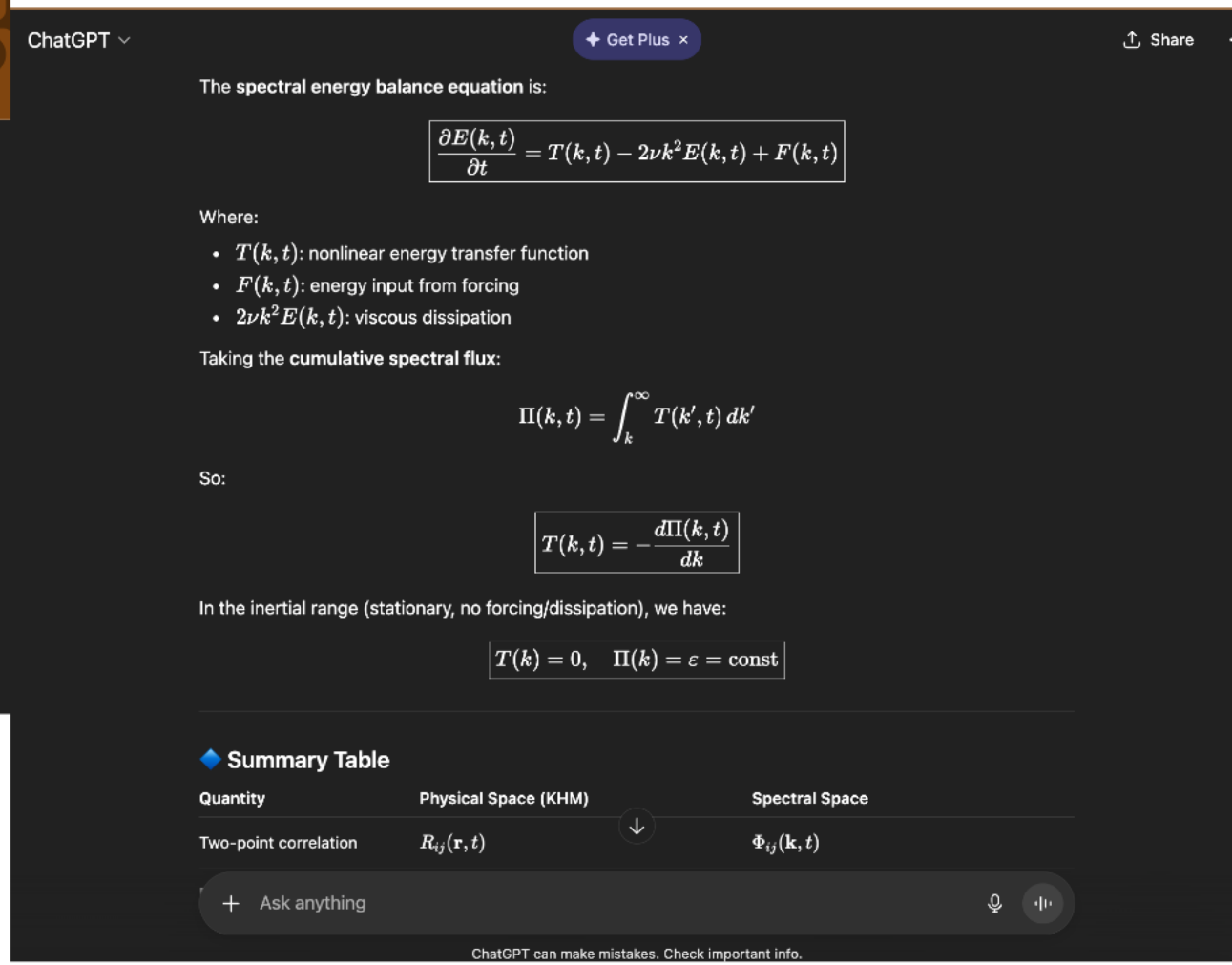
Dhruv Balwada

Physical Oceanographer | Lamont-Doherty Earth Observatory

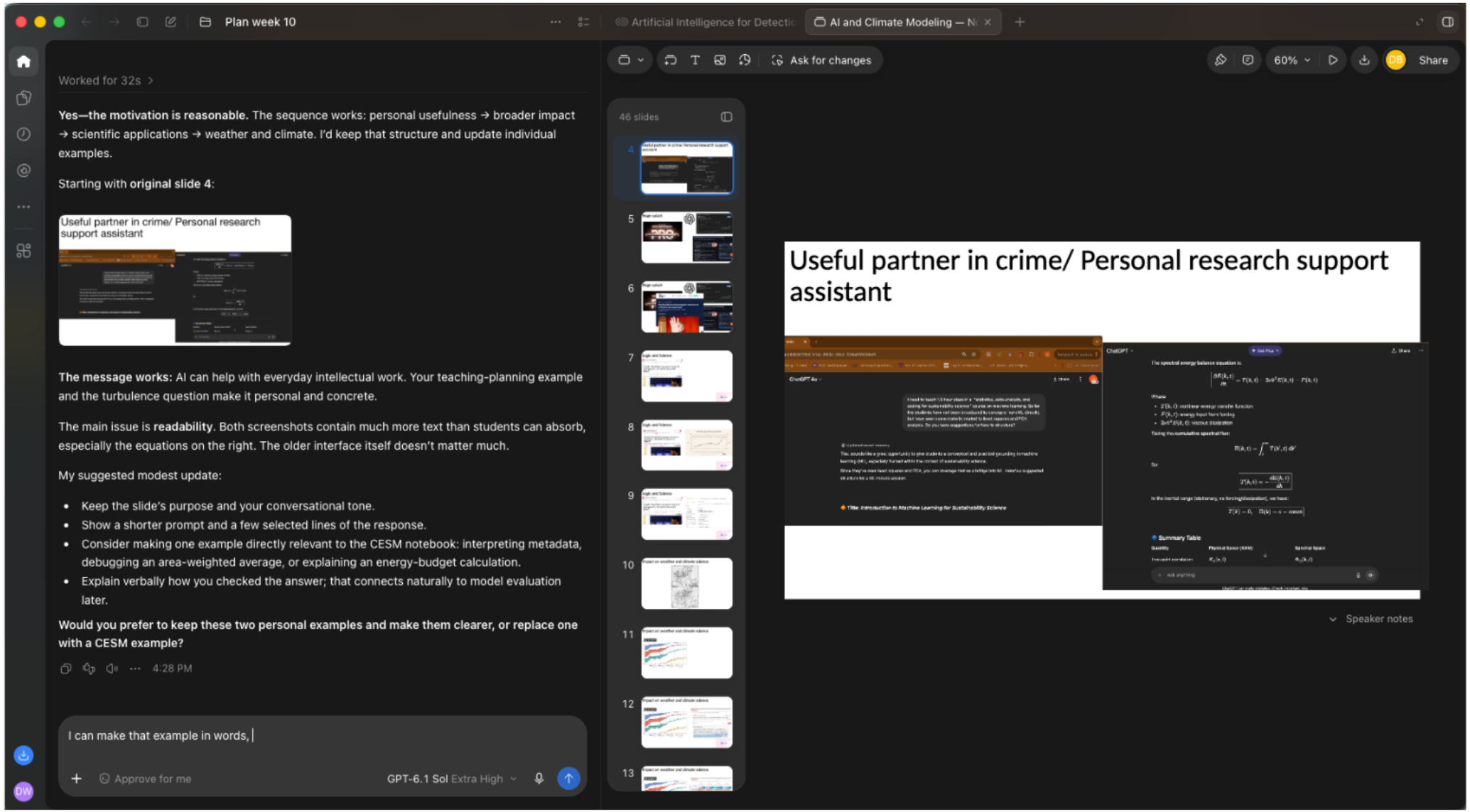

Primer on Machine Learning in Environmental Sciences
Exploring data-driven tools for a changing planet
Dhruv Balwada
Physical Oceanographer | Lamont-Doherty Earth Observatory

Useful partner in crime/ Personal research support assistant





Including these slides...



Huge splash

Everyday use

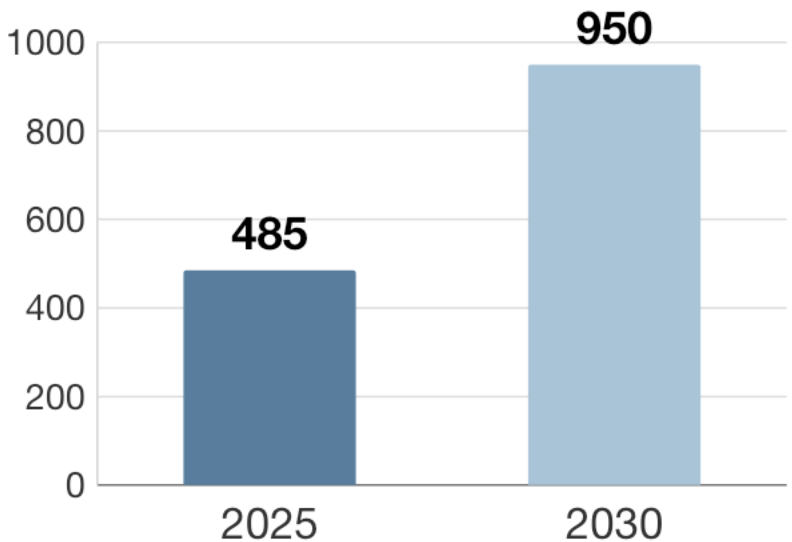
Siri AI across Apple devices
Announcement, June 2026

NVIDIA share price

Five-year history
Screenshot: Nov 11, 2026

Electricity use

Global data centres, TWh per year

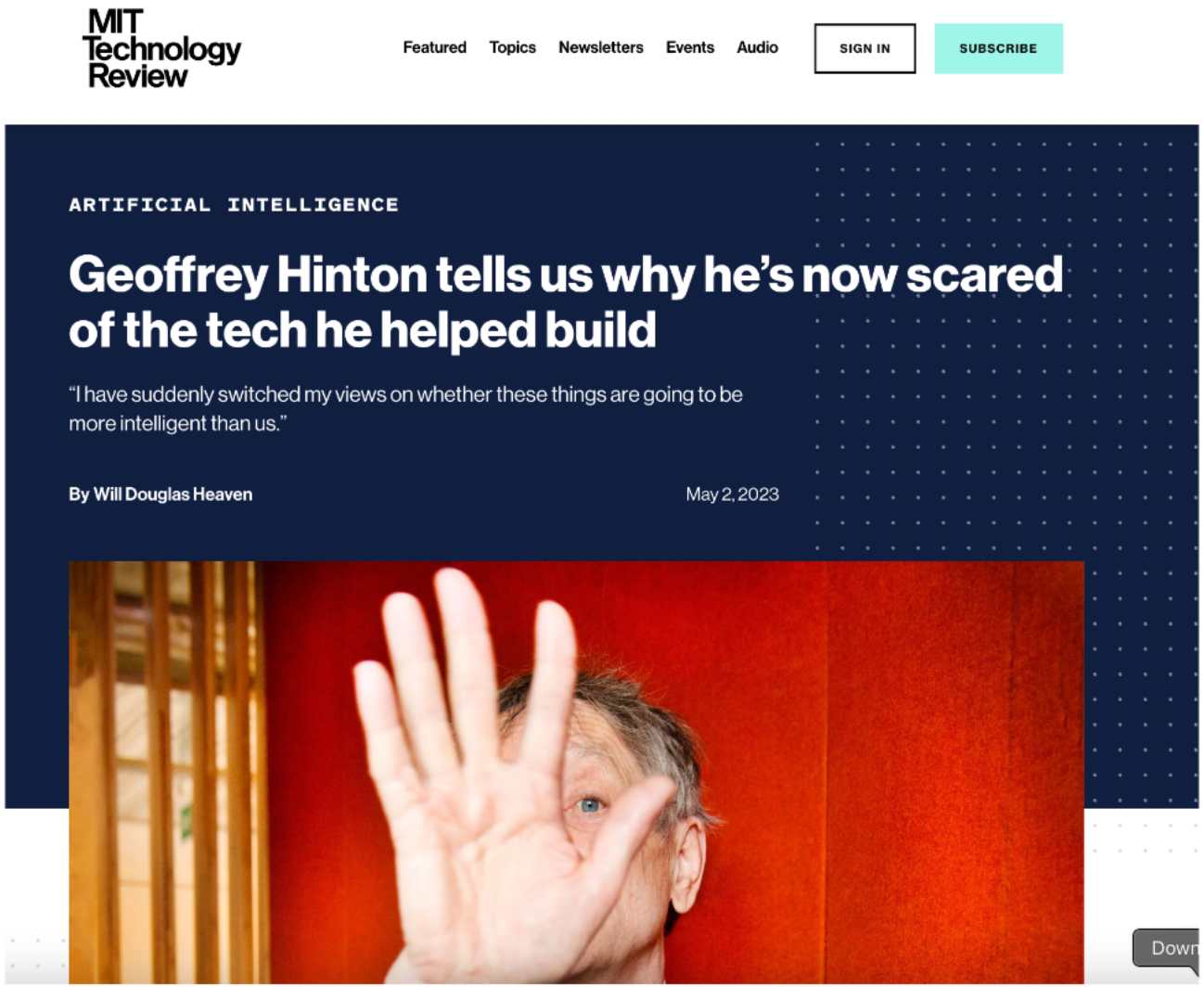


2025 estimate, 2030 projection
All data centres, including AI

Sources: Apple (June 2026), NVIDIA chart screenshot (Nov 11, 2026), IEA (2026)

Huge splash

Historical example: May 2, 2023



Huge splash

Scientific American, September 17, 2026

Read the November Issue: Social Media Addiction

September 17, 2026 | 4 min read Add Us On Google

Does AGI mean what we think it means?

Artificial general intelligence was once a term confined to technical science papers, but its meaning has slipped

BY GEORGE MUSSER EDITED BY CLAIRE CAMERON

Logic and Science

INDEPENDENT

ENHANCED BY Google

NEWS SPORTS VOICES CULTURE LIFESTYLE TRAVEL CLIMATE PREMIUM MORE INDEPENDENT tv

Tech

Google DeepMind computer beats Go champion Lee Se-dol in shock 4-1 victory

Some had thought that Mr Lee's clawing back of the fourth game could indicate that he was learning how the computer worked – but the final game ended in a victory for AlphaGo

Andrew Griffin • Tuesday 15 March 2016 07:18 EDT • 0 Comments

2016

DeepMind

Logic and Science

INDEPENDENT

ENHANCED BY Google

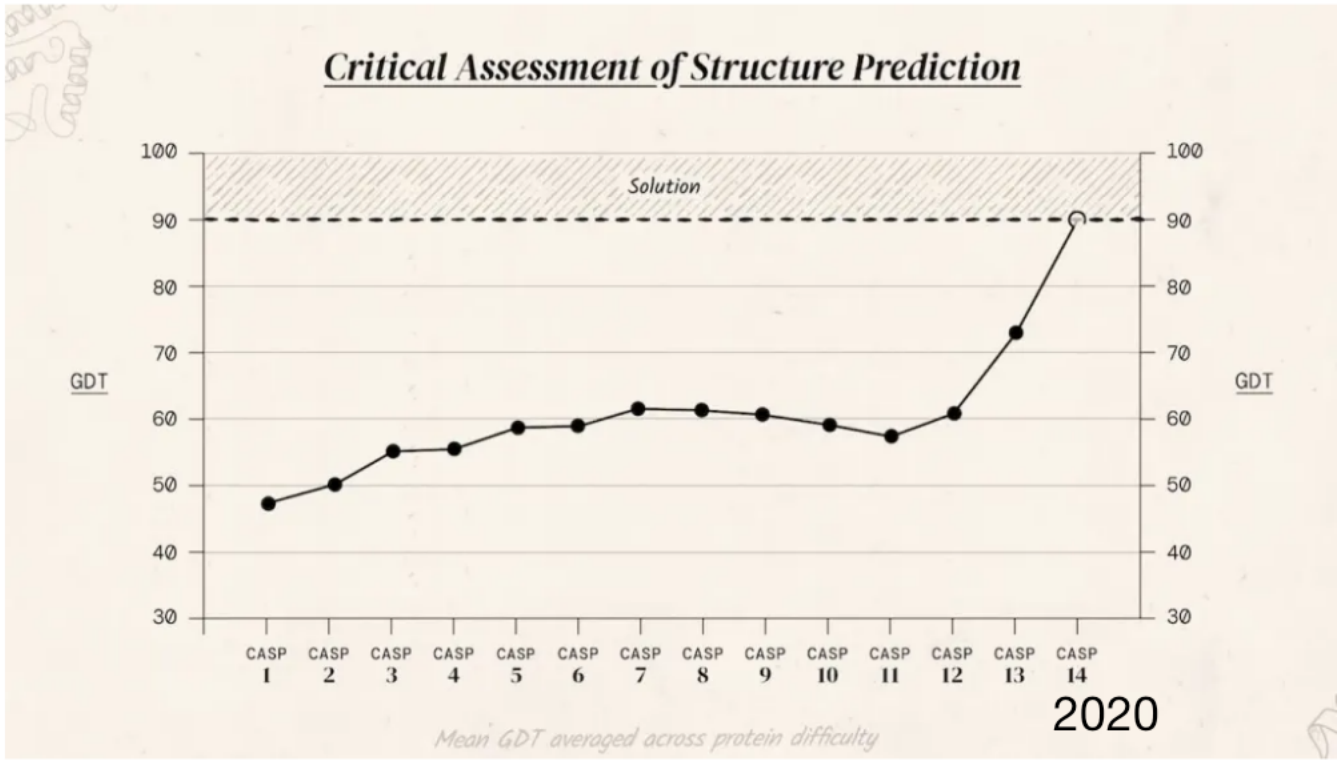
NEWS SPORTS VOICES CULTURE LIFESTYLE TRAVEL CLIMATE PREMIUM MORE INDEPENDENT tv

Tech

Google DeepMind computer beats Go champion Lee Se-dol in shock 4-1 victory

Some had thought that Mr Lee's clawing back of the fourth game could indicate that he was learning how the computer worked – but the final game ended in a victory for AlphaGo

Andrew Griffin • Tuesday 15 March 2016 07:18 EDT • 0 Comments



DeepMind

Logic and Science

INDEPENDENT

ENHANCED BY Google

NEWS SPORTS VOICES CULTURE LIFESTYLE TRAVEL CLIMATE PREMIUM MORE INDEPENDENT tv

Tech

Google DeepMind computer beats Go champion Lee Se-dol in shock 4-1 victory

Some had thought that Mr Lee's clawing back of the fourth game could indicate that he was learning how the computer worked – but the final game ended in a victory for AlphaGo

Andrew Griffin • Tuesday 15 March 2016 07:18 EDT • 0 Comments

THE NOBEL PRIZE

Nobel Prizes & laureates About Stories Educational Events & museums

Nobel Prize in Chemistry 2024

Summary
Laureates
David Baker
Demis Hassabis
John Jumper
Prize announcement
Press release
Popular information
Advanced information
Award ceremony video
Award ceremony speech

Share this

9 October 2024

The Royal Swedish Academy of Sciences has decided to award the Nobel Prize in Chemistry 2024

with one half to

David Baker
University of Washington, Seattle, WA, USA
Howard Hughes Medical Institute, USA

"for computational protein design"

and the other half jointly to

Demis Hassabis
Google DeepMind, London, UK

John Jumper
Google DeepMind, London, UK

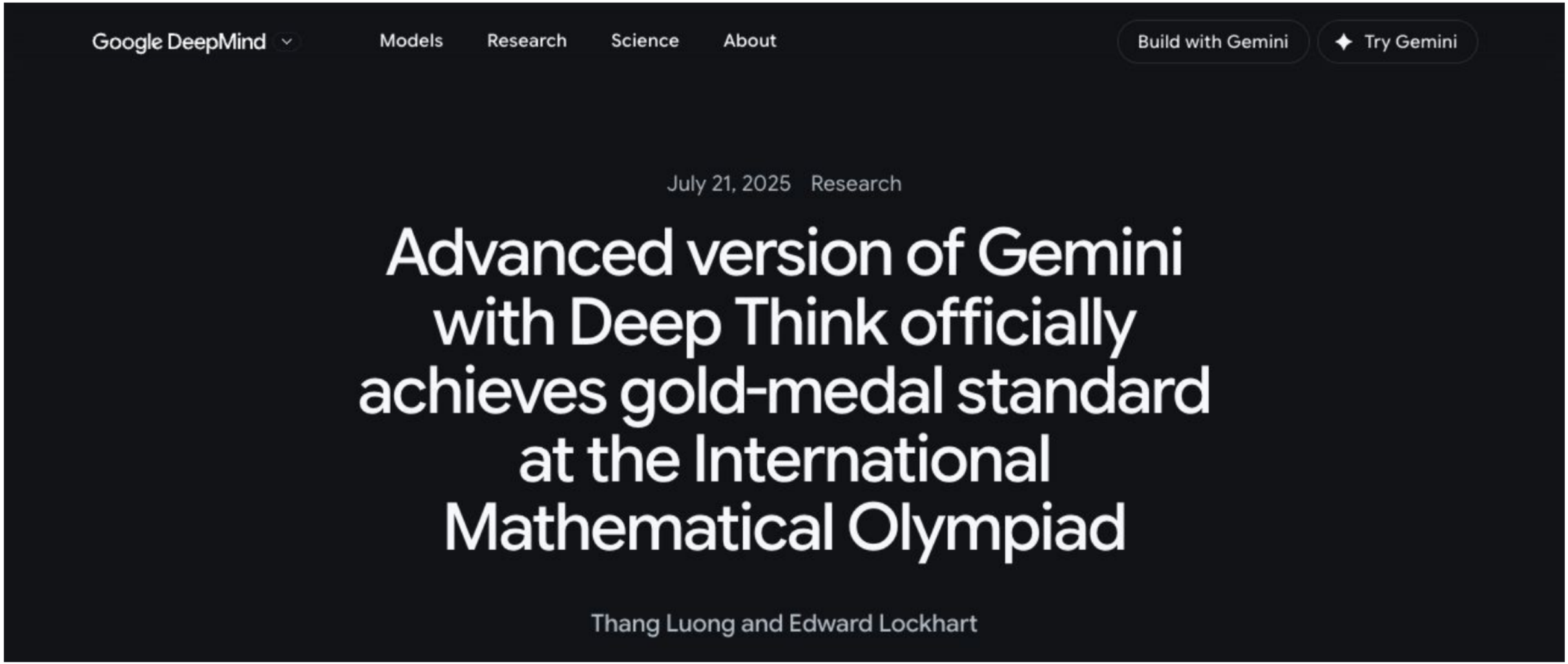
"for protein structure prediction"

2024

DeepMind

Logic and Science

2025



35/42 points. Five of six problems solved. Solutions graded by IMO coordinators.

Logic and Science

June 2026

Numerically observed relation (2014)
Analytical proof with Claude's help (2026)

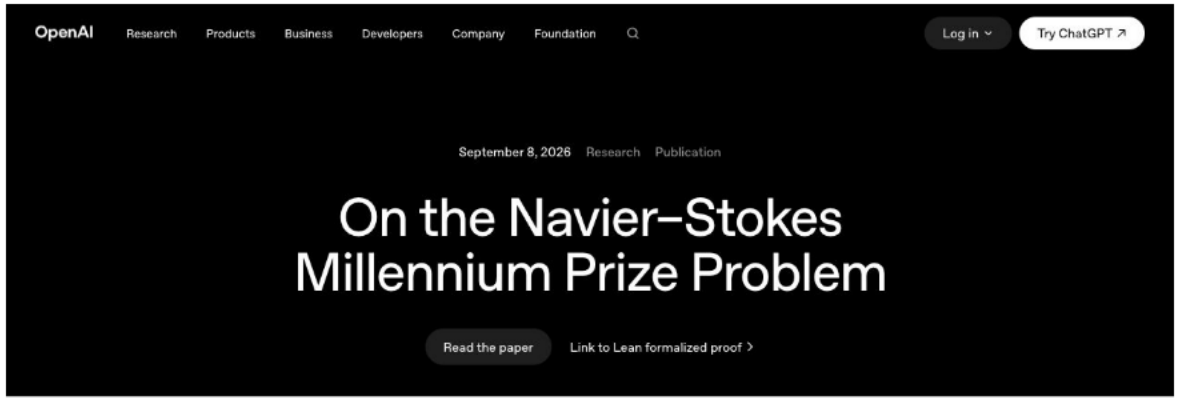
Parisi and Zamponi verified the proof.

Parisi & Zamponi, Journal of Statistical Mechanics (2026) 073301

Logic and Science

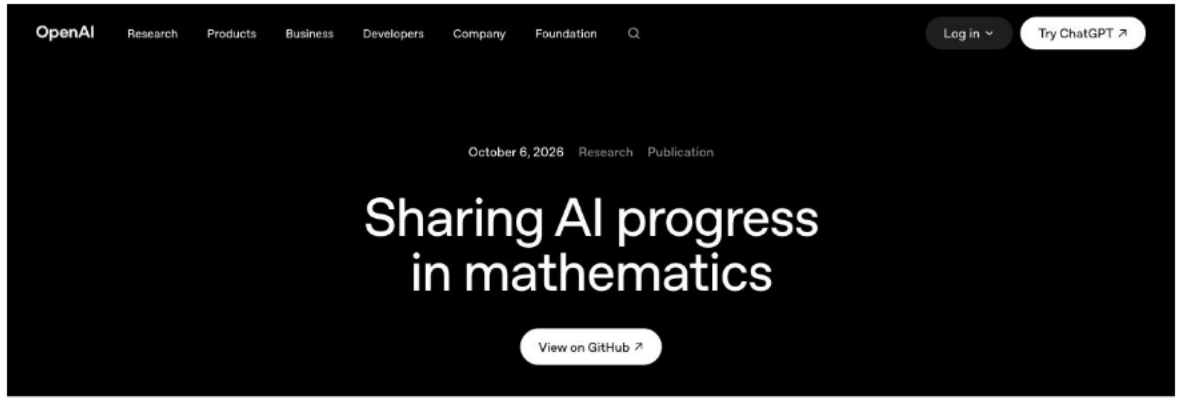
2026

September 2026



Announced finite-time blowup with smooth forcing

October 2026

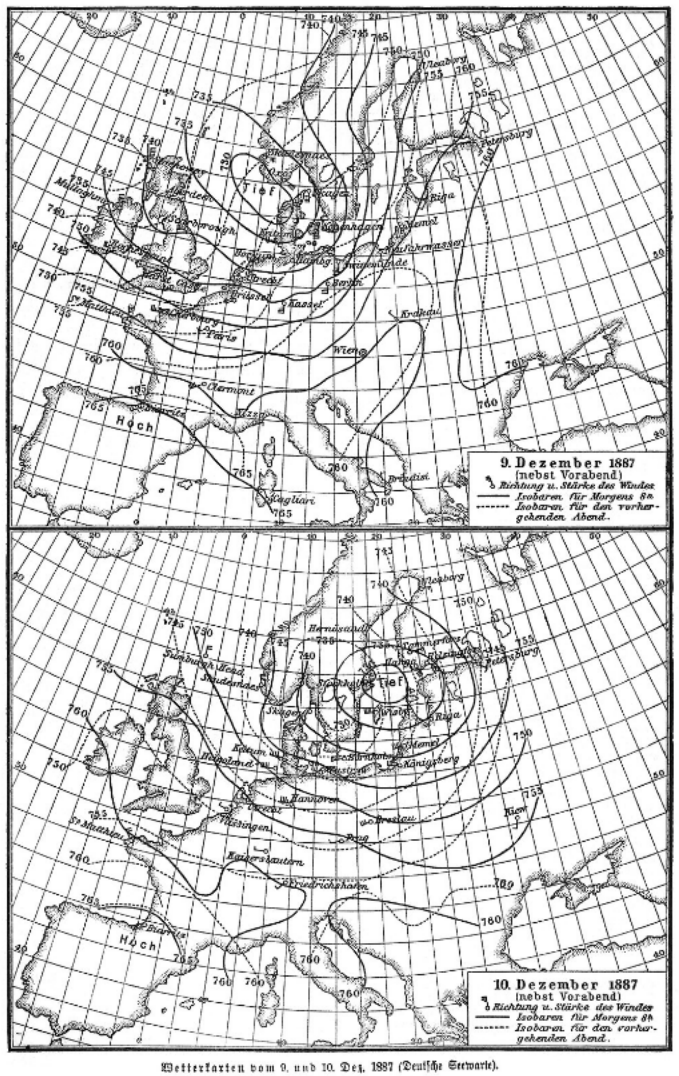


Research manuscripts and formalized proofs

Navier–Stokes: announced result under evaluation as of November 11, 2026.

Impact on weather and climate science

Historical weather charts (1887)



Wetterkarten vom 9. und 10. Dez. 1887 (Deutsche Seewarte).

Analog forecasting

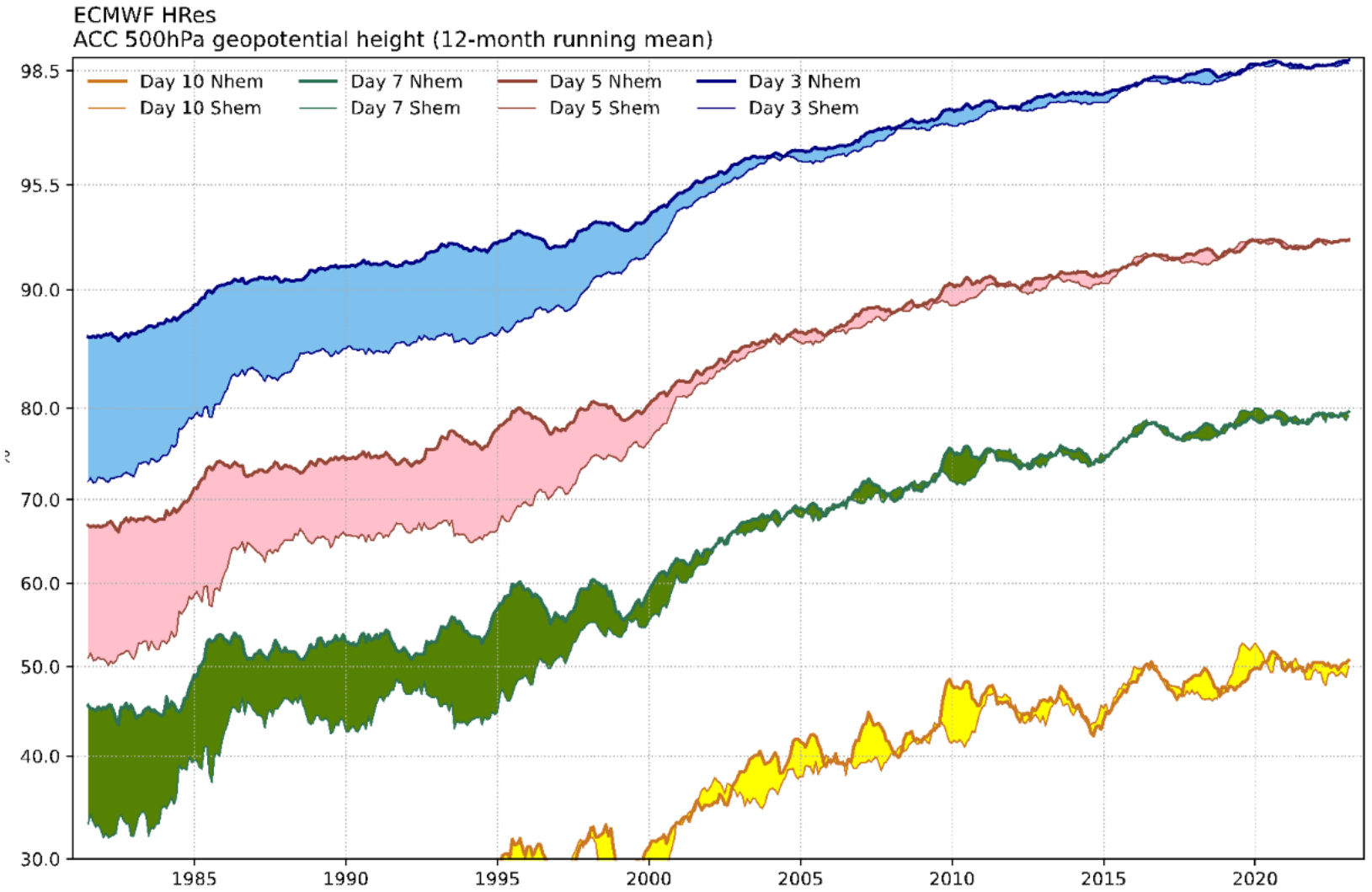
Find similar weather patterns in the historical record.
Use what happened next as a forecast.

Pressure (2026)

Analog forecasting
in the D-Day debate

Impact on weather and climate science

ECMWF 1981–2023: 500 hPa height anomaly correlation (%)



Numerical weather prediction

1922: Richardson's book
1950: computer forecasts

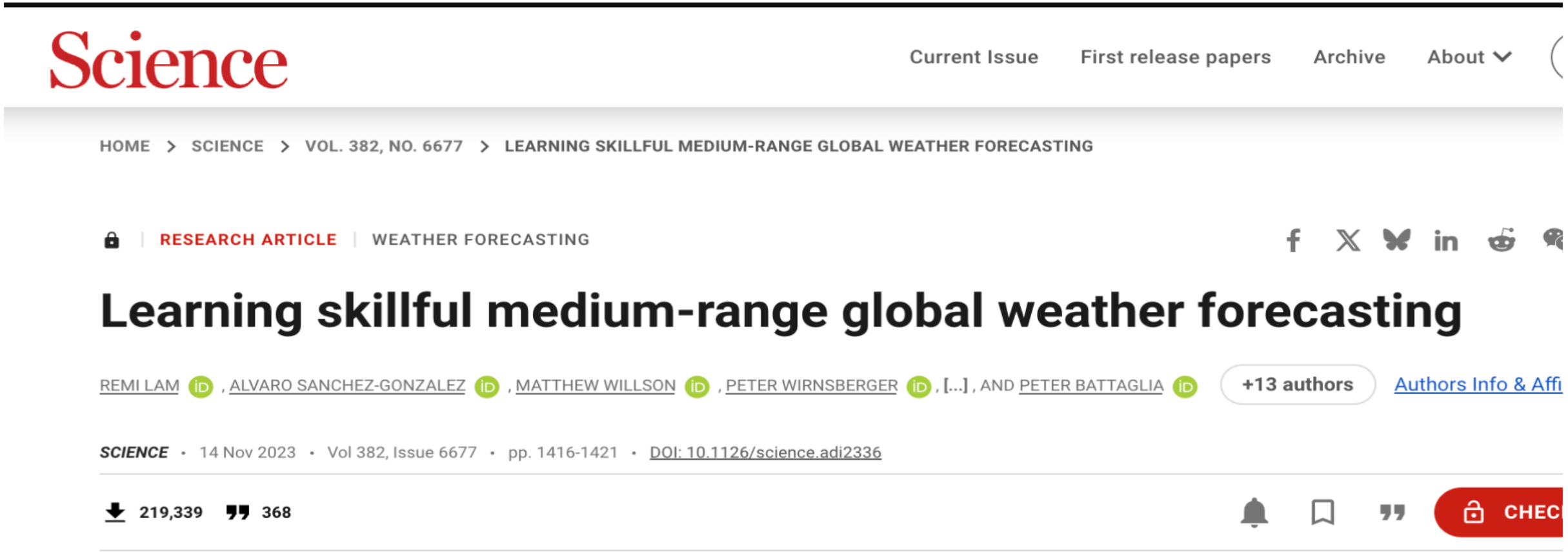
Better observations and data assimilation

Better physics and numerics

More computing power and finer resolution

Impact on weather and climate science

GraphCast (2023)



Science

Current Issue First release papers Archive About

HOME > SCIENCE > VOL. 382, NO. 6677 > LEARNING SKILLFUL MEDIUM-RANGE GLOBAL WEATHER FORECASTING

RESEARCH ARTICLE | WEATHER FORECASTING

Learning skillful medium-range global weather forecasting

REMI LAM, ALVARO SANCHEZ-GONZALEZ, MATTHEW WILLSON, PETER WIRNSBERGER, [...], AND PETER BATTAGLIA +13 authors Authors Info & Affi

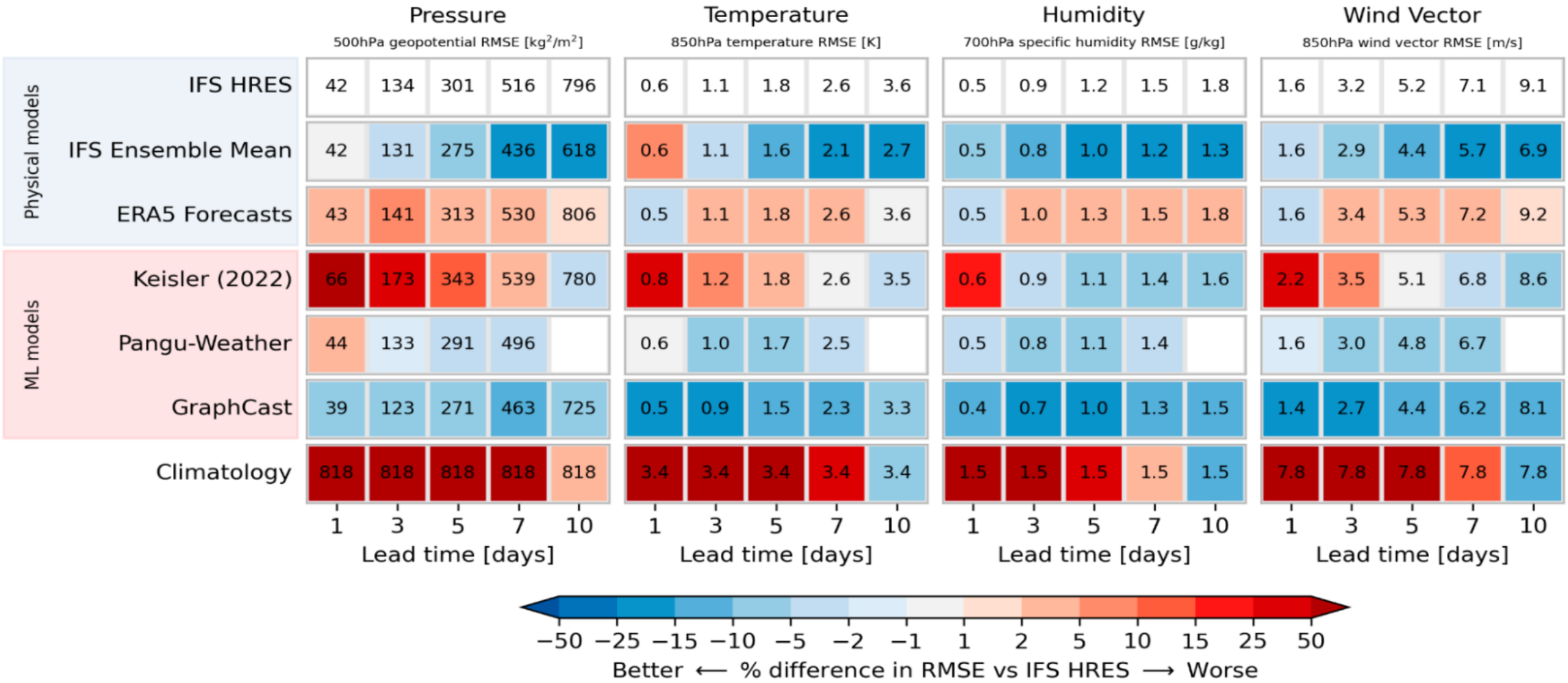
SCIENCE • 14 Nov 2023 • Vol 382, Issue 6677 • pp. 1416-1421 • DOI: 10.1126/science.adi2336

219,339 368 CHEC

Trained on ERA5: observations combined with a physical model
Learns the next atmospheric state, then repeats that forecast step

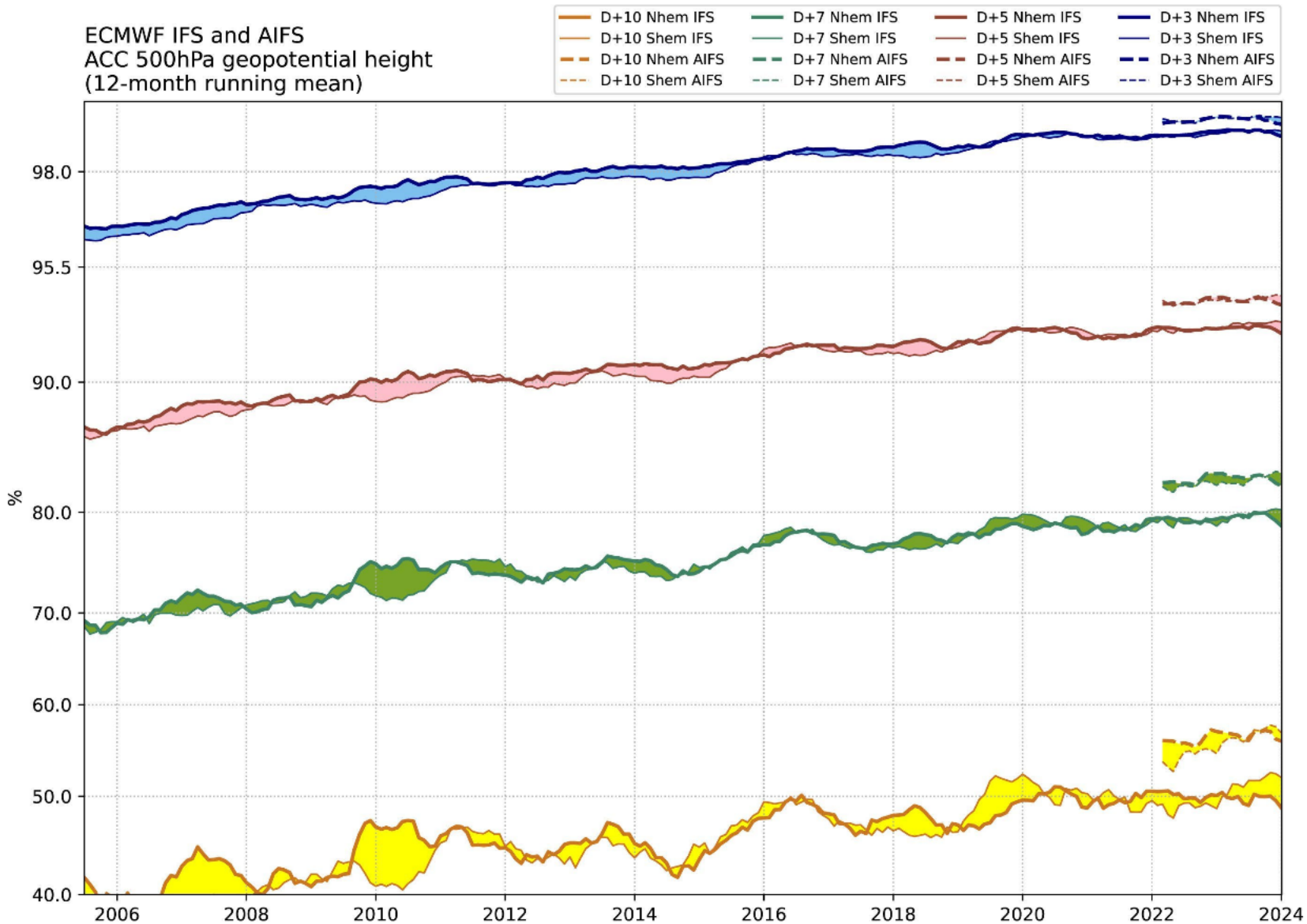
Impact on weather and climate science

GraphCast: the original benchmark (2023)



Blue: lower RMSE relative to IFS HRES. Lam et al., Science (2023)

Impact on weather and climate science



AIFS and IFS

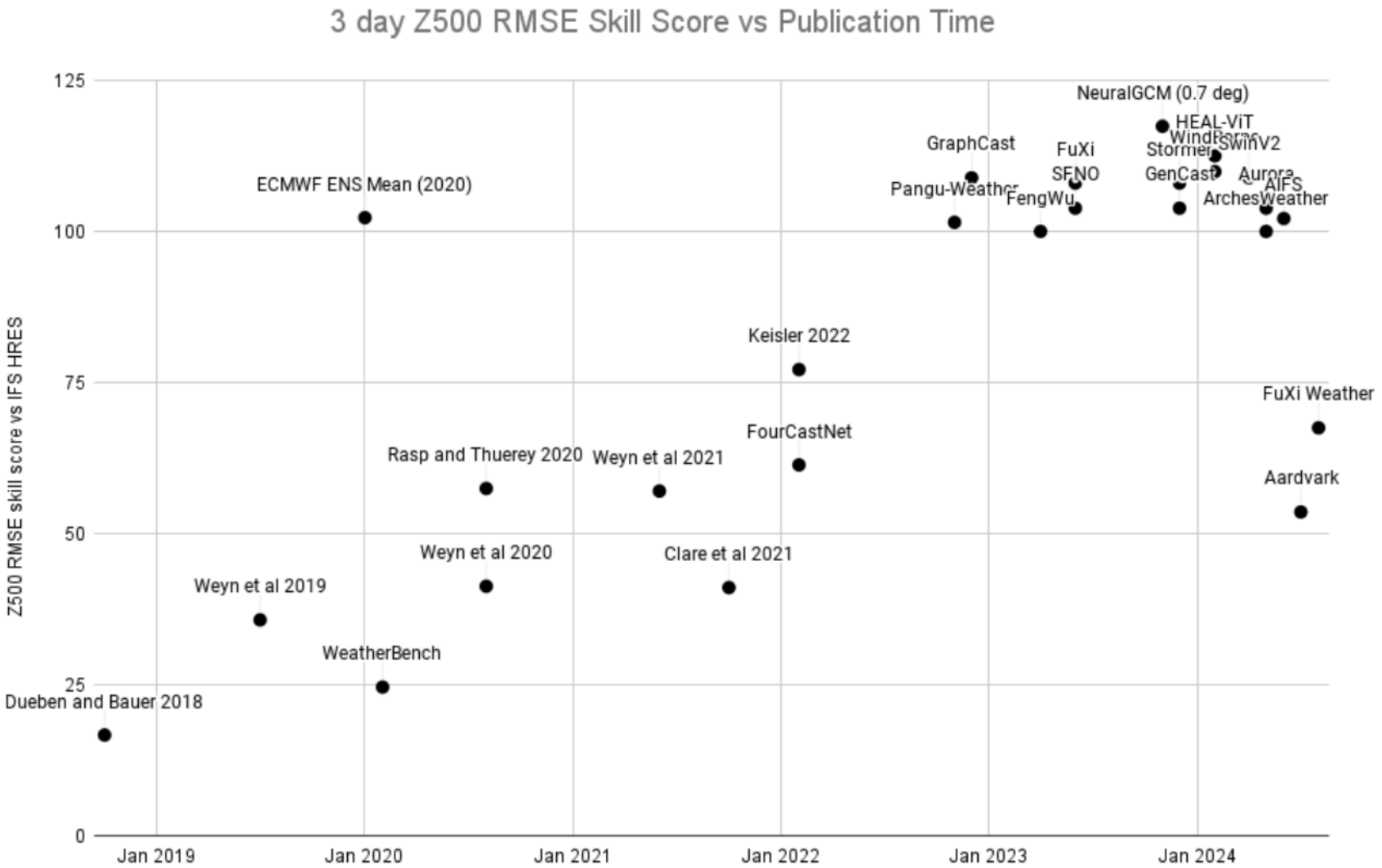
Solid: physics-based IFS
Dashed: AI-based AIFS

Higher correlation at the same forecast lead time

Retrospective tests from 2022
Operational AIFS began in 2025

ECMWF, Dueben (2025)

Progress in AI weather forecasting



3-day Z500 forecasts

100 = reference IFS HRES
Above 100 = lower RMSE

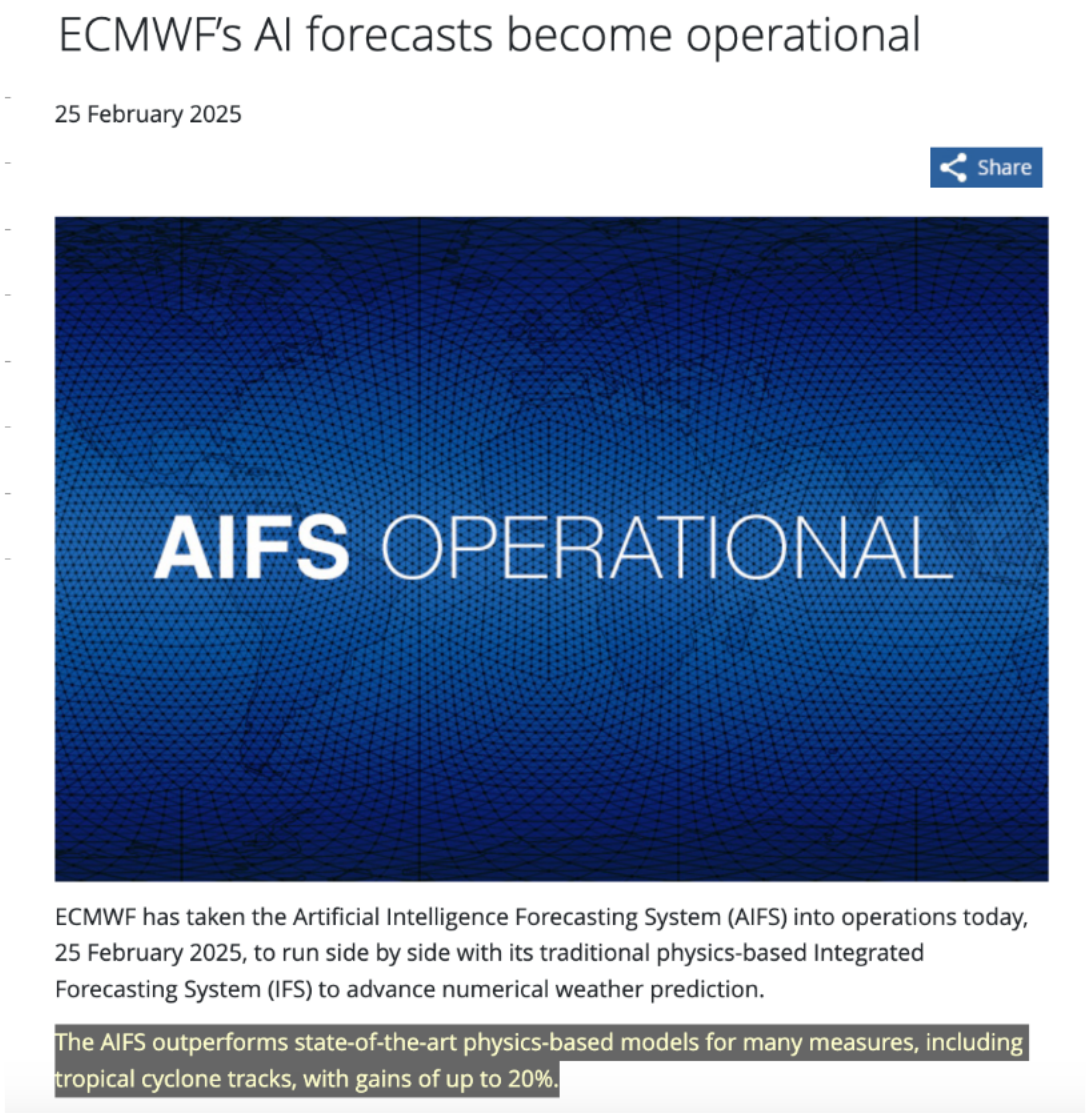
Rapid catch-up from 2018 to 2022–23

Historical compilation through 2024

Approximate comparisons: evaluation setups differ

Source: Stephan Rasp (2024)

Impact on weather and climate science



ECMWF's AI forecasts become operational

25 February 2025

Share

ECMWF has taken the Artificial Intelligence Forecasting System (AIFS) into operations today, 25 February 2025, to run side by side with its traditional physics-based Integrated Forecasting System (IFS) to advance numerical weather prediction.

The AIFS outperforms state-of-the-art physics-based models for many measures, including tropical cyclone tracks, with gains of up to 20%.

25 February 2025

AIFS Single becomes operational

1 July 2025

AIFS ensemble becomes operational

12 May 2026

Version 2 updates to both systems

Runs alongside physics-based IFS

Impact on weather and climate science

ECMWF's AI forecasts become operational

25 February 2025

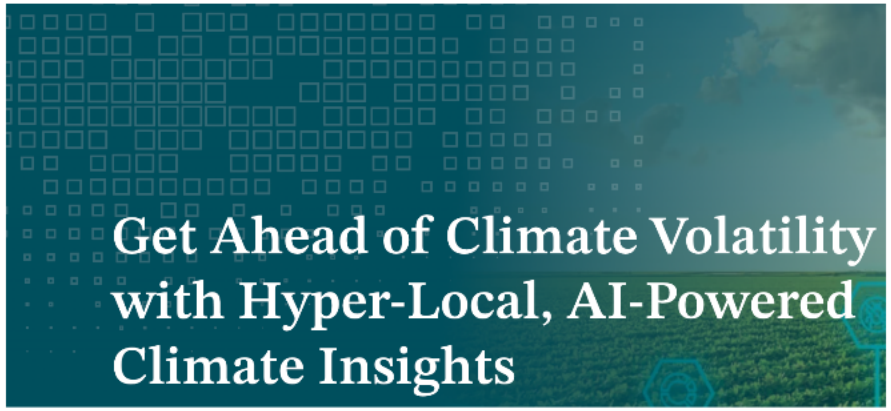
Share

ECMWF has taken the Artificial Intelligence Forecasting System (AIFS) into operations today, 25 February 2025, to run side by side with its traditional physics-based Integrated Forecasting System (IFS) to advance numerical weather prediction.

The AIFS outperforms state-of-the-art physics-based models for many measures, including tropical cyclone tracks, with gains of up to 20%.

ClimateAi

Industries Solutions & Products Decision Accuracy



Impact on weather and climate science

ECMWF's AI forecasts become operational

25 February 2025

Share

ECMWF has taken the Artificial Intelligence Forecasting System (AIFS) into operations today, 25 February 2025, to run side by side with its traditional physics-based Integrated Forecasting System (IFS) to advance numerical weather prediction.

The AIFS outperforms state-of-the-art physics-based models for many measures, including tropical cyclone tracks, with gains of up to 20%.

ClimateAi

Industries Solutions & Products Decision Accuracy

Ai2

Open technologies + Research + About Us Blog Playground

AI for the environment

Climate modeling

Since the early days of climate modeling, software, hardware, and the ways that engineers and scientists collaborate have gone through incredible transformations. Better data and technologies will inform how we mitigate and adapt to global impacts, such as sea level rise, community destruction, and biodiversity loss.

Impact on weather and climate science

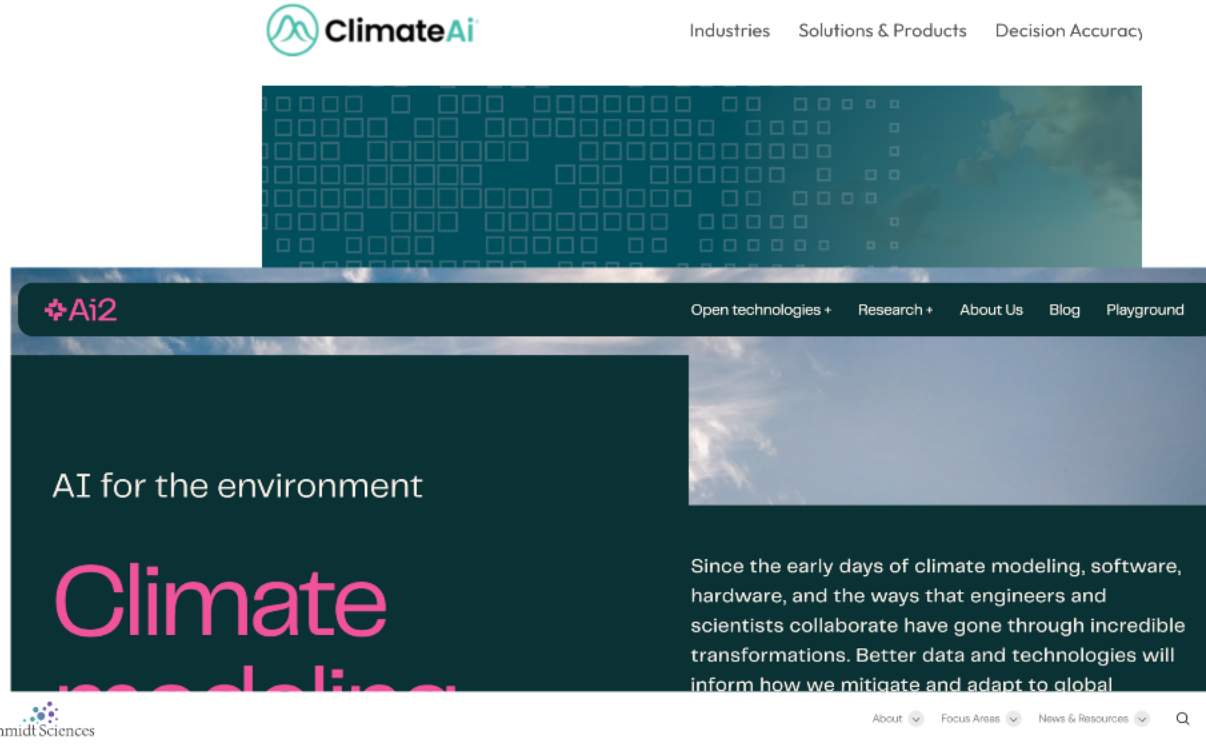
ECMWF's AI forecasts become operational

25 February 2025

Share

ECMWF has taken the Artificial Intelligence Forecasting System (AIFS) into operations today, 25 February 2025, to run side by side with its traditional physics-based Integrated Forecasting System (IFS) to advance numerical weather prediction.

The AIFS outperforms state-of-the-art physics-based models for many measures, including tropical cyclone tracks, with gains of up to 20%.



What is ML?

🧠 Framing ML: Not Magic — Just Pattern Recognition & Function Approximation

> "At its core, machine learning isn't magic — it's just a way of learning patterns from data, and using those patterns to make predictions or decisions."

✨ Key Message:

> **ML is the art of finding a function that maps inputs to outputs using data — not by deriving equations from first principles, but by learning them from examples.**

What is ML?

🔍 Pattern Recognition

- Humans do this naturally: given enough examples, we learn to distinguish cloudy from clear skies, or recognize a familiar sound.
- ML systems formalize this process:
  - Given input features (e.g., temperature, humidity, wind)
  - And observed outputs (e.g., rainfall tomorrow)
  - Learn the patterns that link them

> "In supervised ML, the computer is shown examples of inputs and correct outputs — and it learns the patterns that link them."

📐 Function Approximation

- In math terms, ML builds an approximation to a function:

$$y = f(x) + \epsilon$$

  where:

  - x: input features
  - y: target/output
  - f: function learned from data
  - ϵ: noise or irreducible error

- This is the same mindset as least squares fitting — but with more flexible, often nonlinear function forms.

> "The model doesn't *know* the true equations of motion — it just finds a function that seems to work well based on the data it sees."

What is ML?

🔍 Pattern Recognition

- Humans do this naturally: given enough examples, we learn to distinguish cloudy from clear skies, or recognize a familiar sound.
- ML systems formalize this process:
  - Given input features (e.g., temperature, humidity, wind)
  - And observed outputs (e.g., rainfall tomorrow)
  - Learn the patterns that link them

> "In supervised ML, the computer is shown examples of inputs and correct outputs — and it learns the patterns that link them."

📐 Function Approximation

- In math terms, ML builds an approximation to a function:

$$y = f(x) + \epsilon$$

  where:

  - x: input features
  - y: target/output
  - f: function learned from data
  - ϵ: noise or irreducible error
- This is the same mindset as least squares fitting — but with more flexible, often nonlinear function forms.

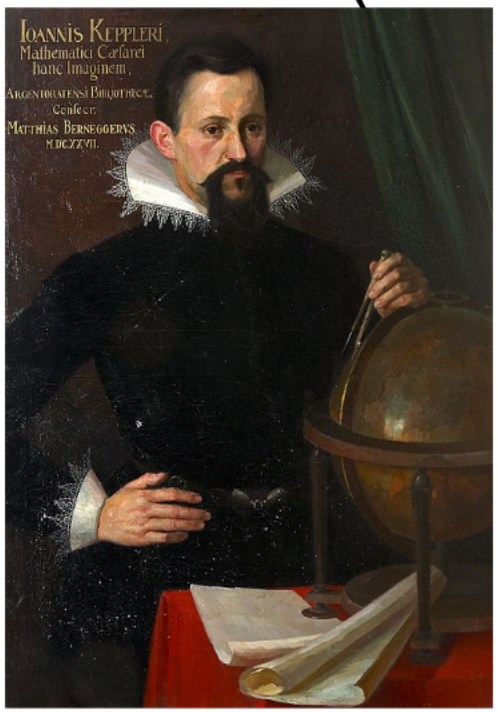
> "The model doesn't *know* the true equations of motion — it just finds a function that seems to work well based on the data it sees."

What kind of a function is a large language model?
- Next word = f (previous words)

Function approximation from data (human learning)

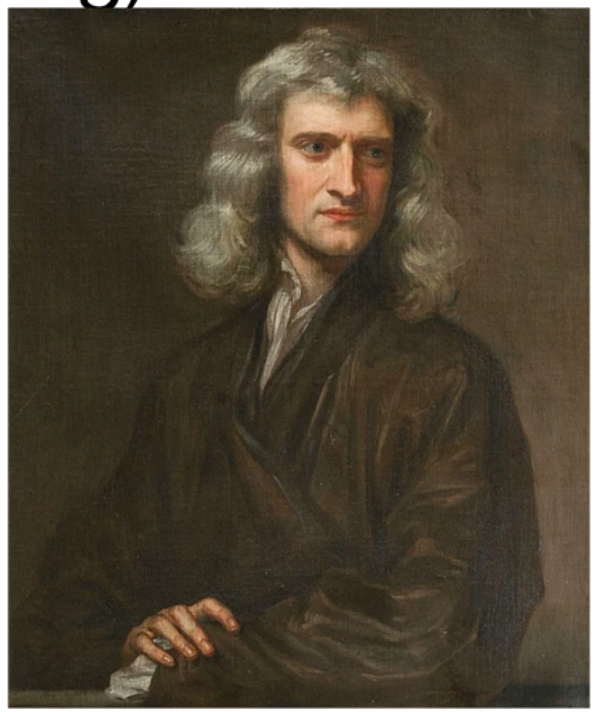
Tycho Brahe (1546-1601)

DATA



Johannes Kepler (1571-1630)

REFINED ANALYSIS



Isaac Newton (1643-1727)

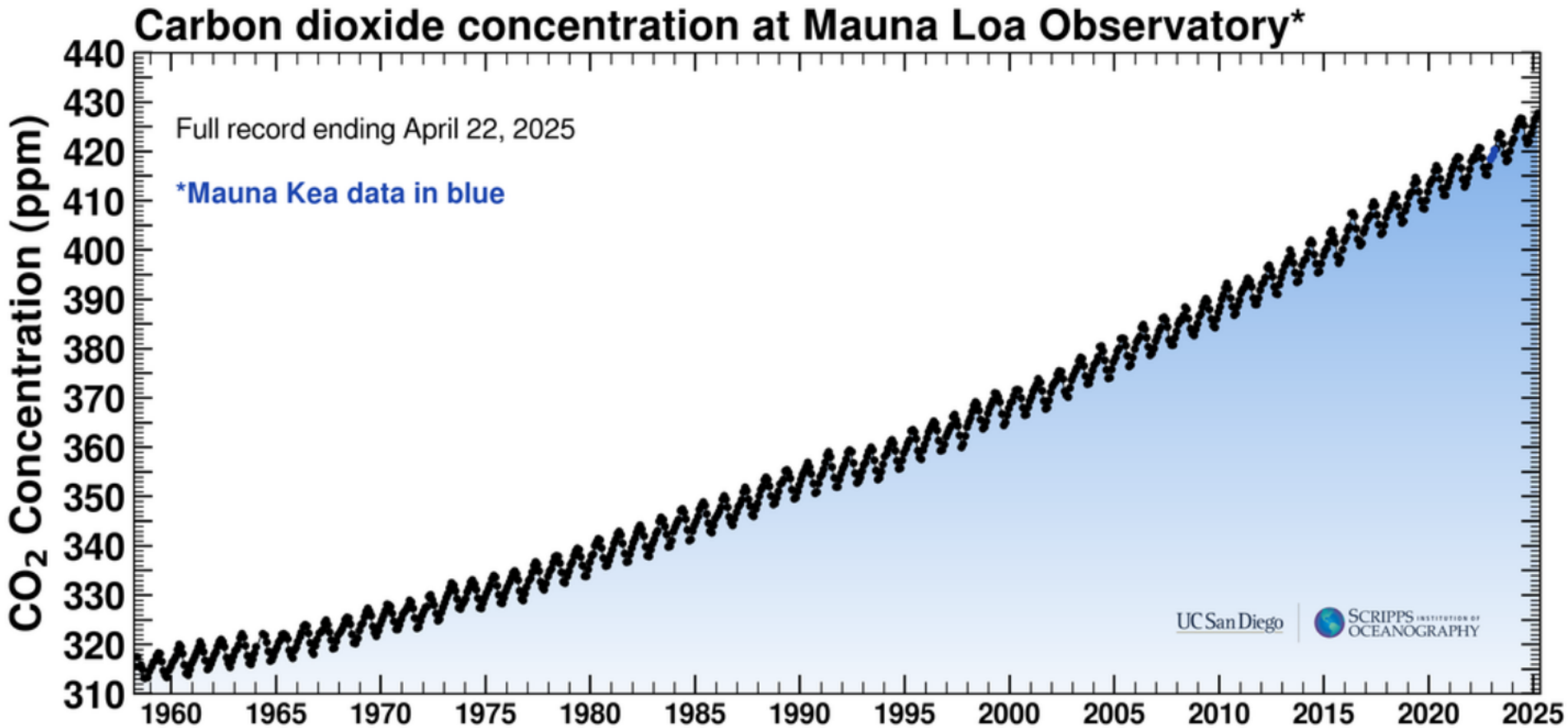
GENERAL PRINCIPLES

Newton's law of gravitation:

$$F = G\frac{m_1 m_2}{r^2} = f(m_1, m_2, r|G)$$

Function. 3 inputs. 1 parameter.

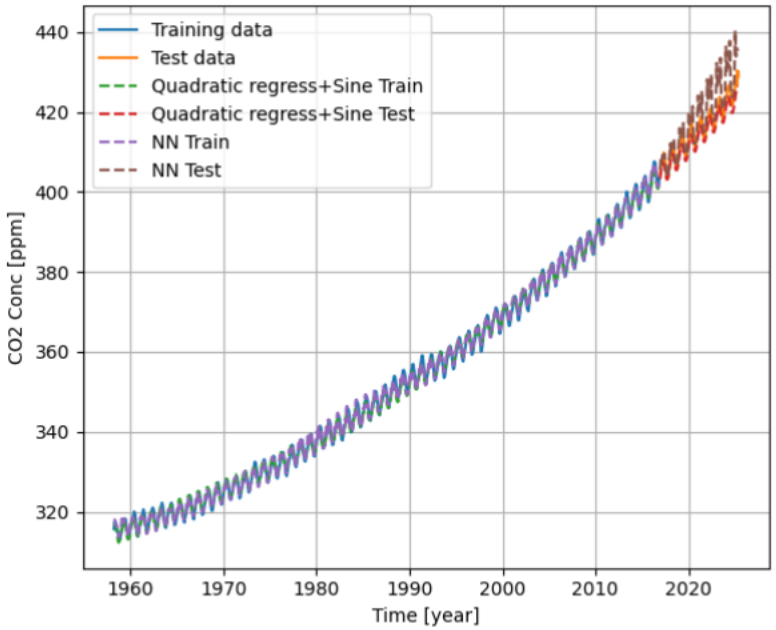
Function approximation from data

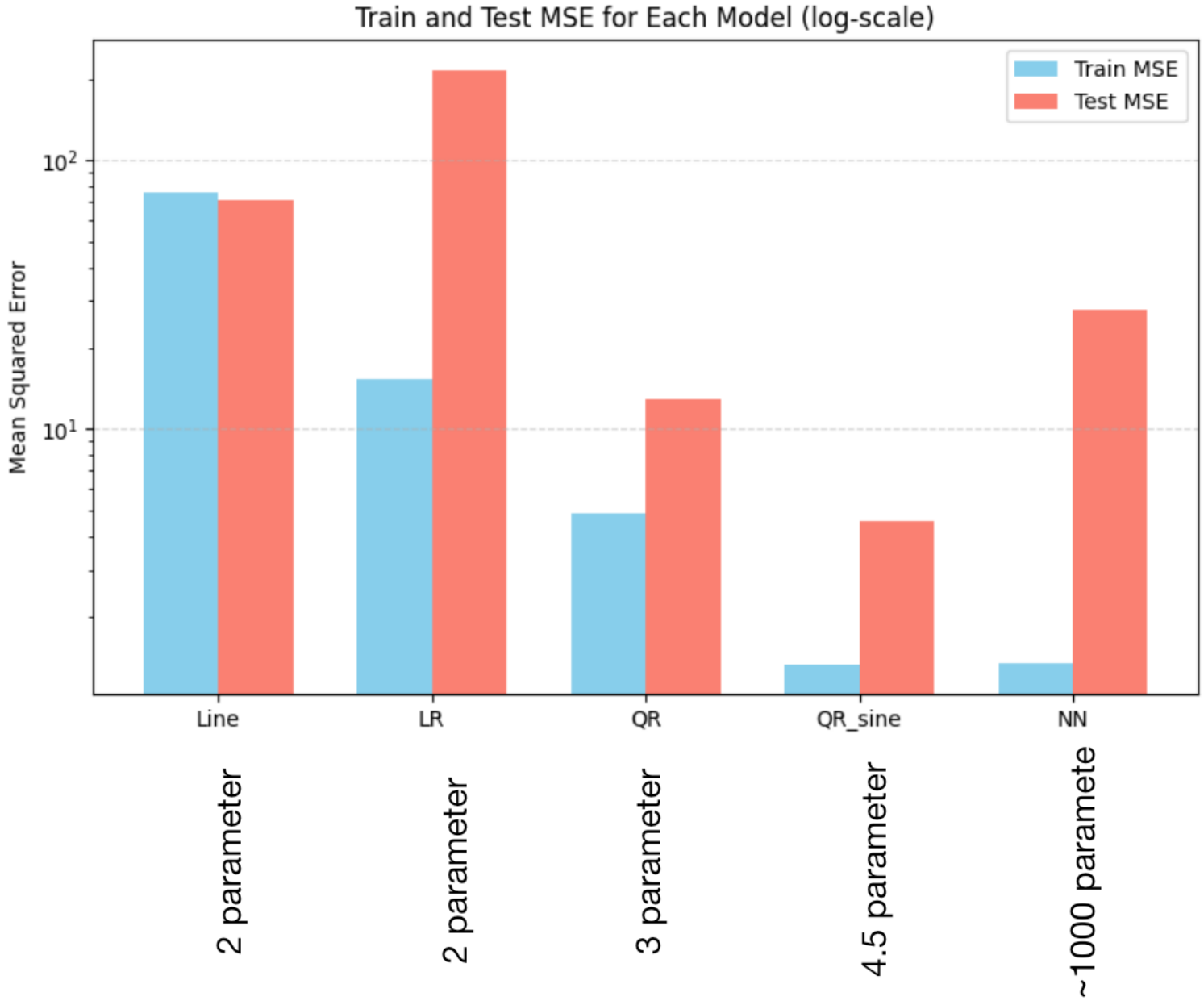


Function approximation from data

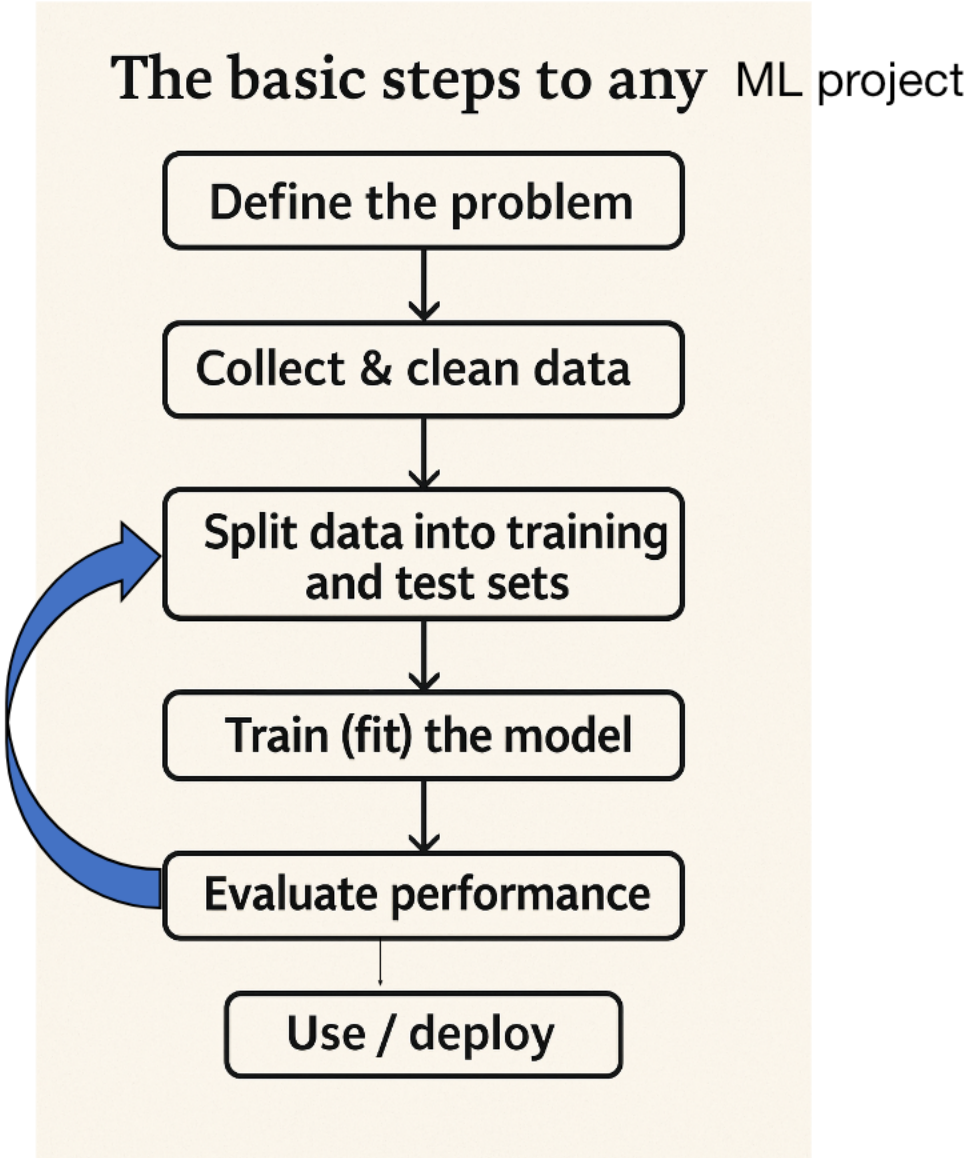
[Class notebook: Fitting the Keeling curve (Labs, Week 10)]

Function approximation from data

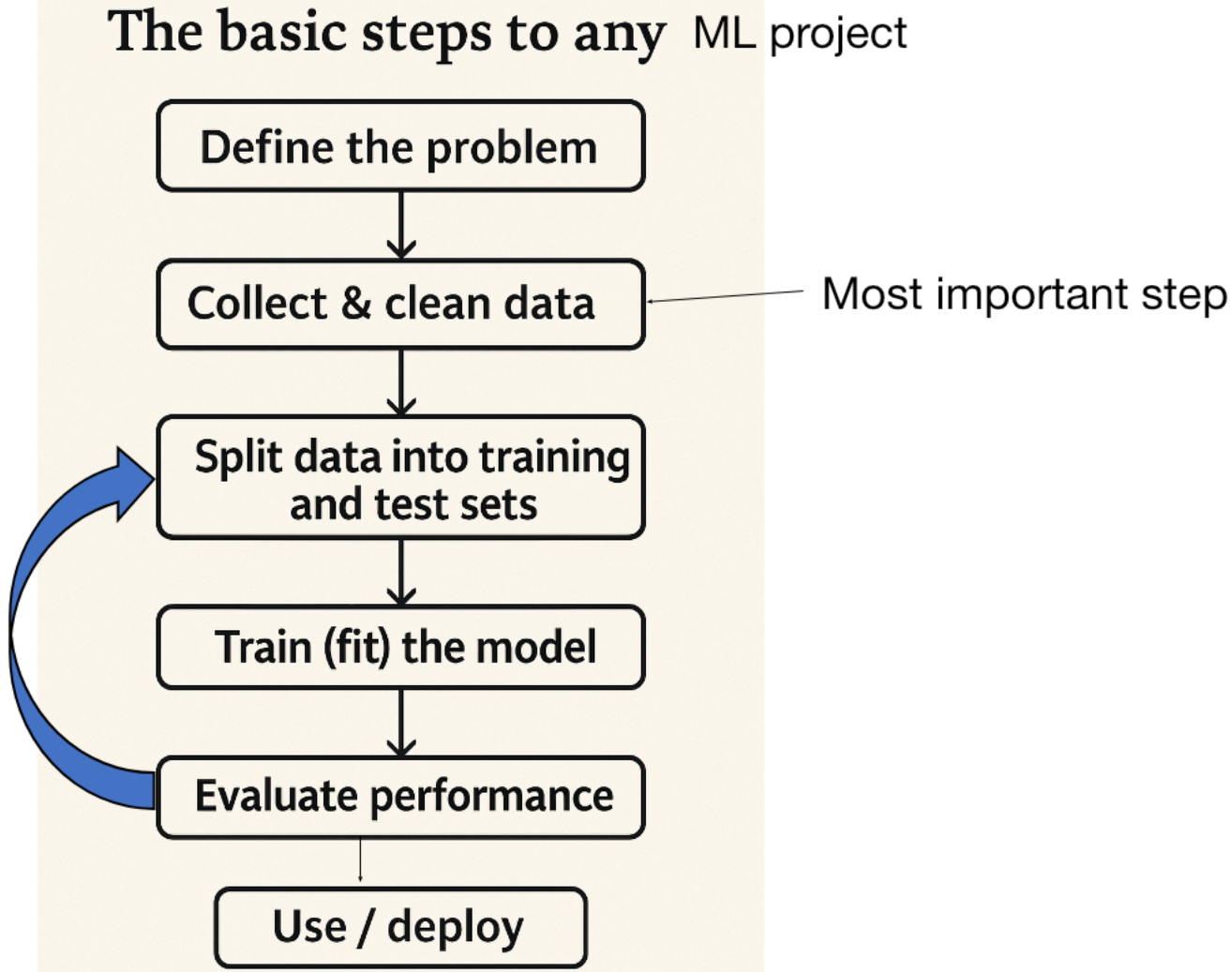




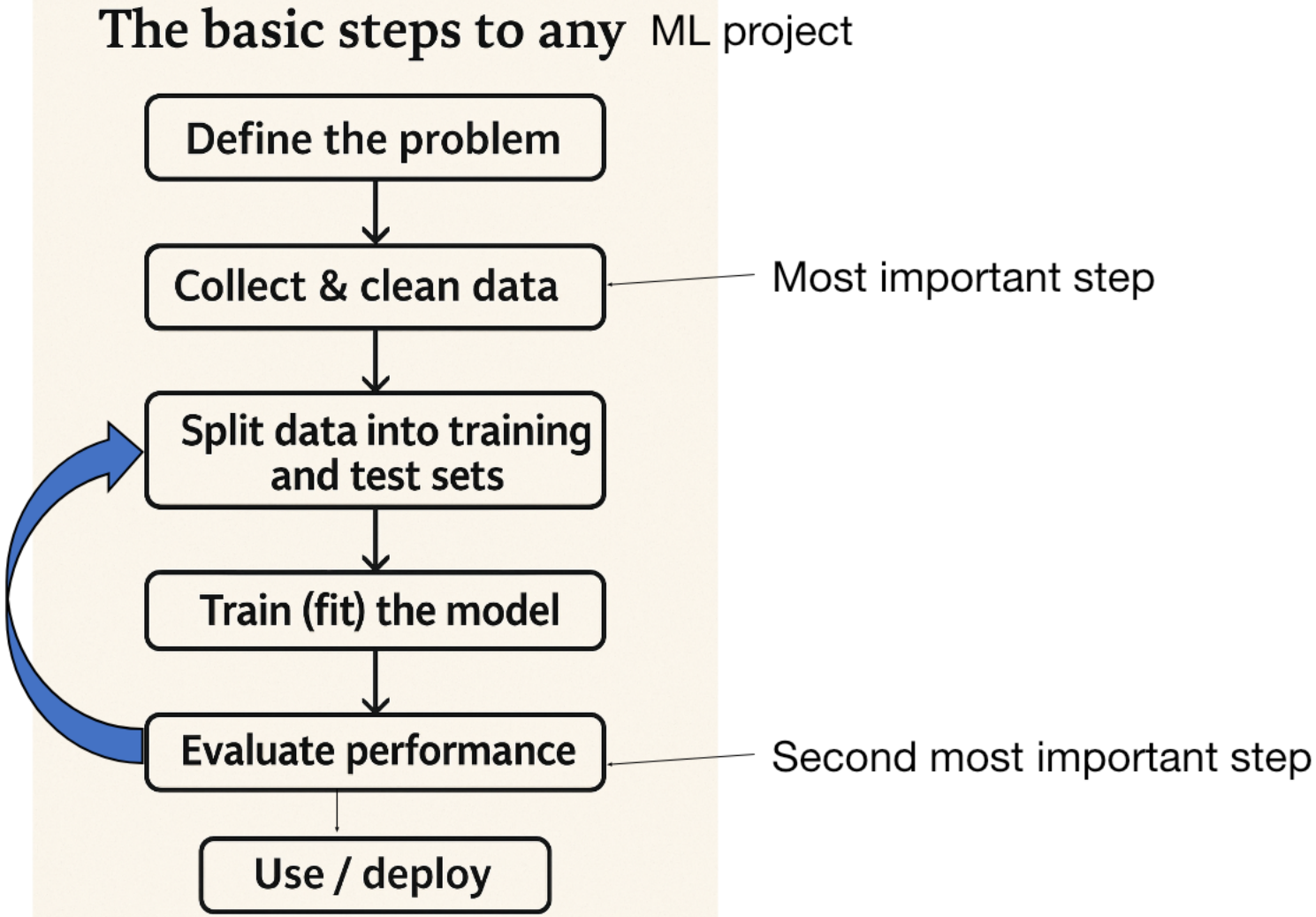
ML Workflow


The basic steps to any ML project
Define the problem
Collect & clean data
Split data into training and test sets
Train (fit) the model
Evaluate performance
Use / deploy

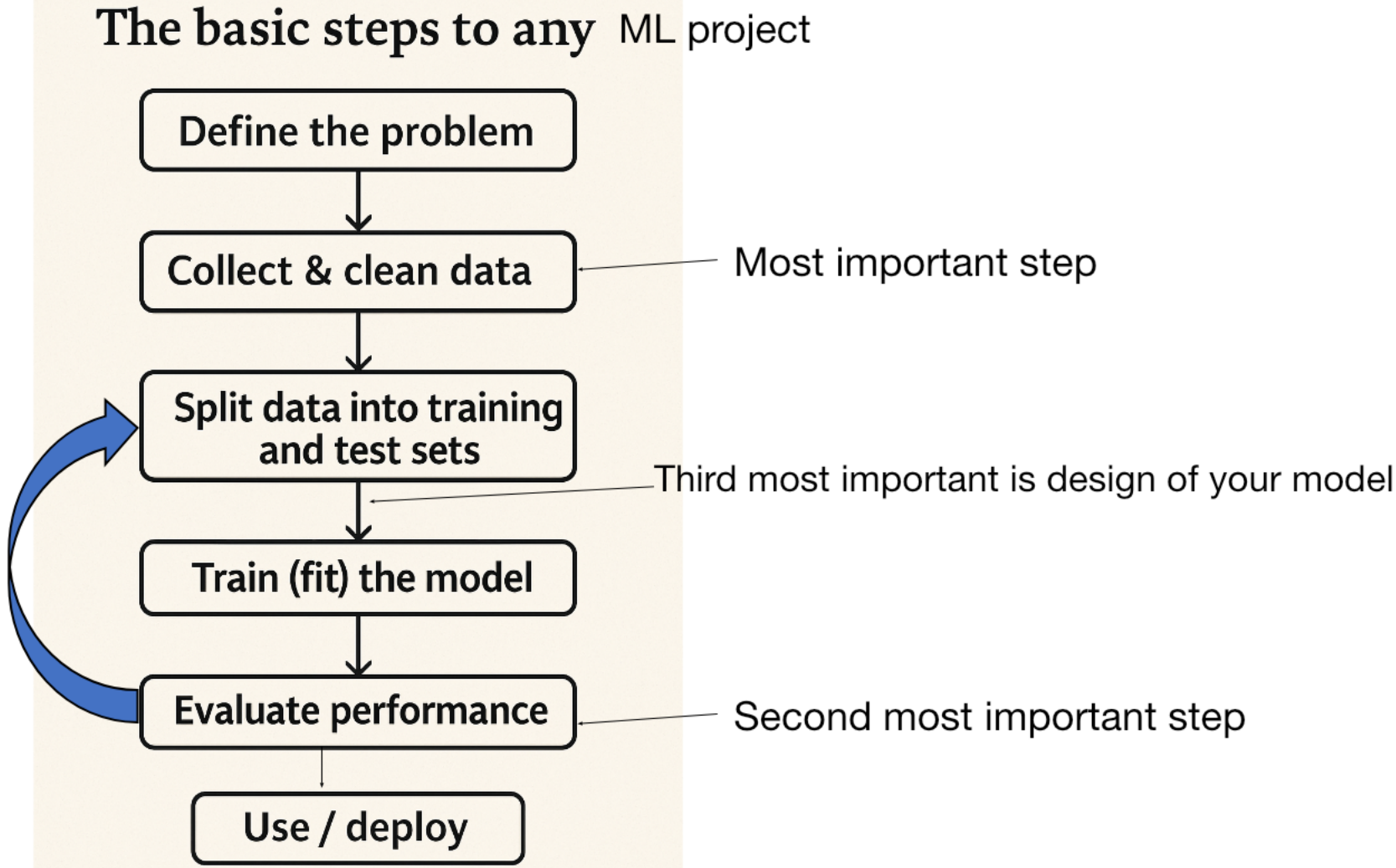
ML Workflow


The basic steps to any ML project
Define the problem
Collect & clean data
Most important step
Split data into training and test sets
Train (fit) the model
Evaluate performance
Use / deploy

ML Workflow


The basic steps to any ML project
Define the problem
Collect & clean data
Most important step
Split data into training and test sets
Train (fit) the model
Evaluate performance
Second most important step
Use / deploy

ML Workflow


The basic steps to any ML project
Define the problem
Collect & clean data
Most important step
Split data into training and test sets
Third most important is design of your model
Train (fit) the model
Evaluate performance
Second most important step
Use / deploy

ML in atmospheric modeling

Forecast emulation

$$\text{future state} \approx f(\text{recent states}; \theta)$$

Example: predict the atmosphere six hours ahead

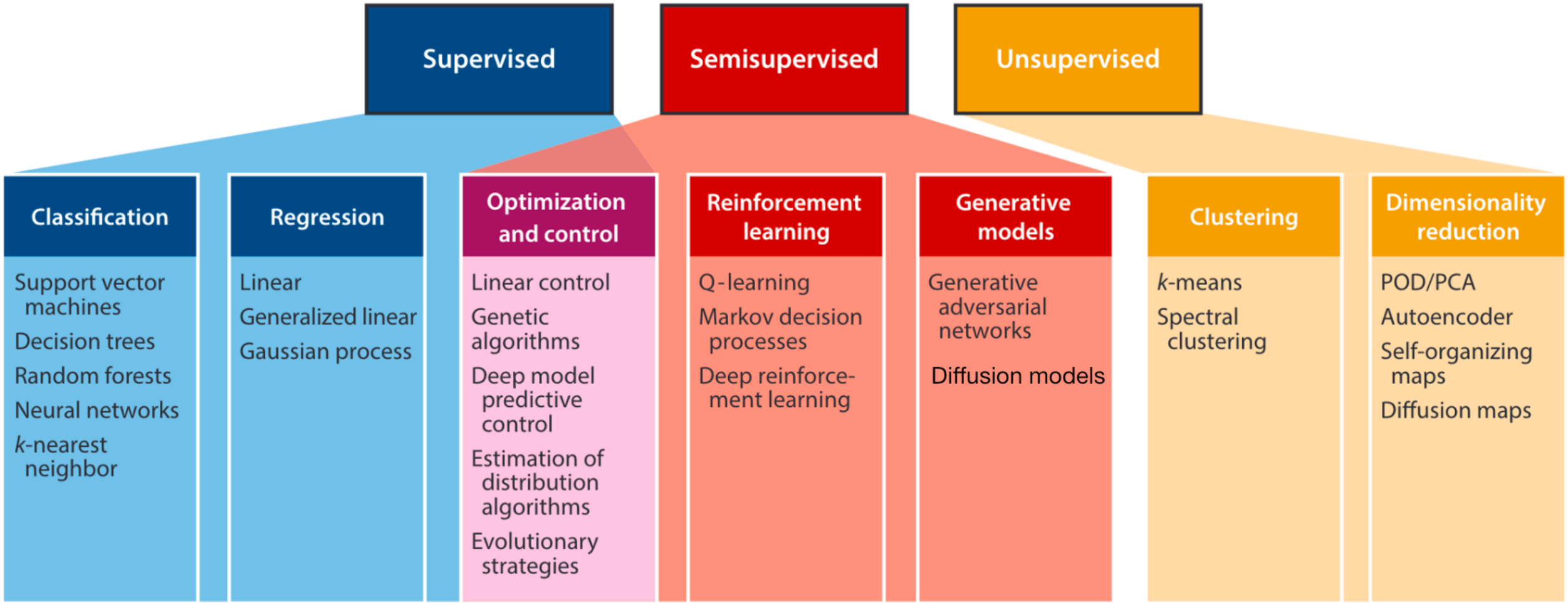
Learned parameterizations

Resolved atmospheric conditions predict unresolved heating and moistening

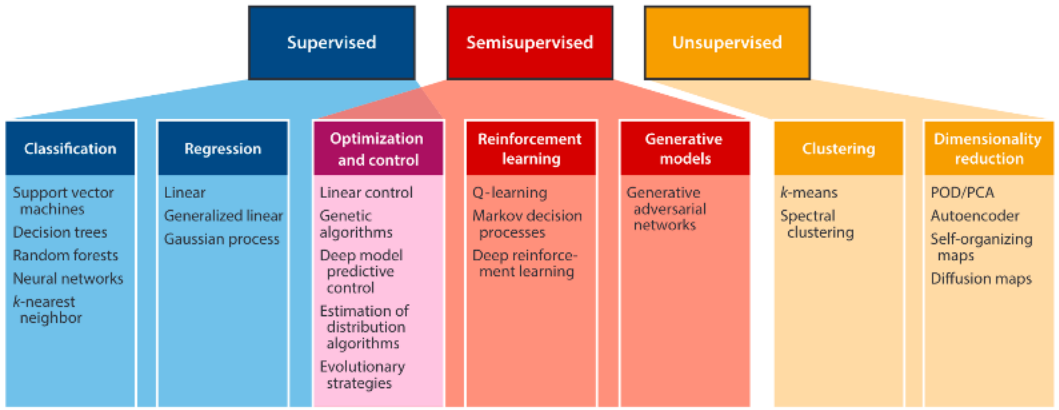
Forecast correction

Model forecasts and local context predict corrected weather variables

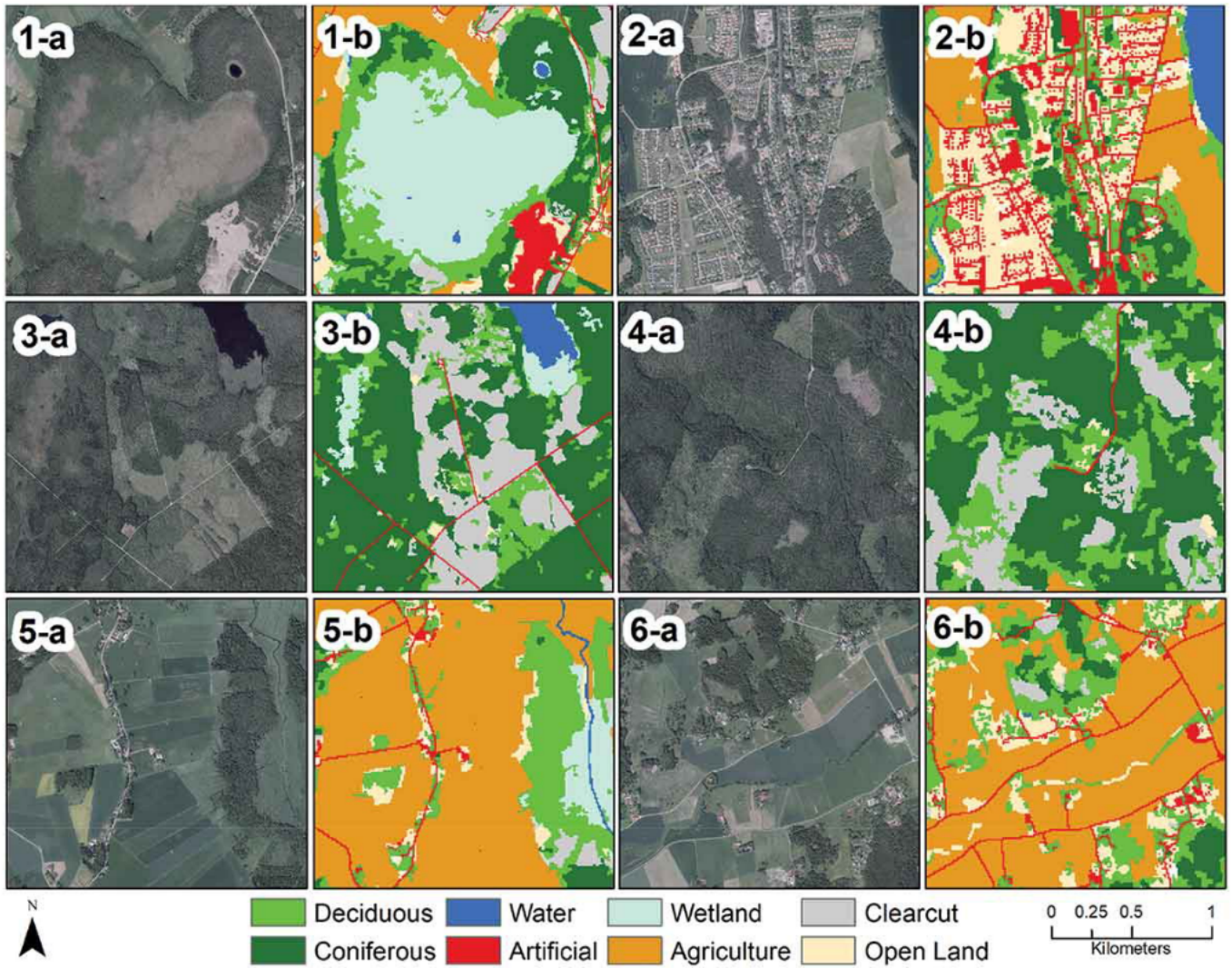
Different types of “machine learning”



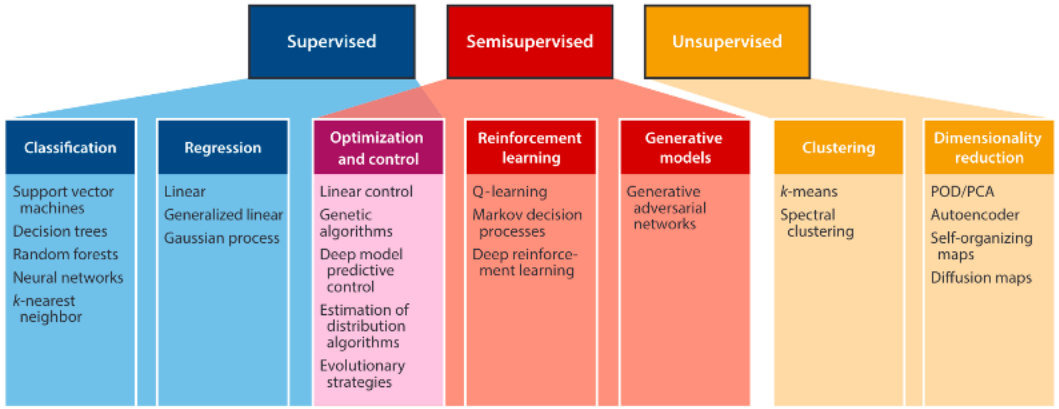
Classification

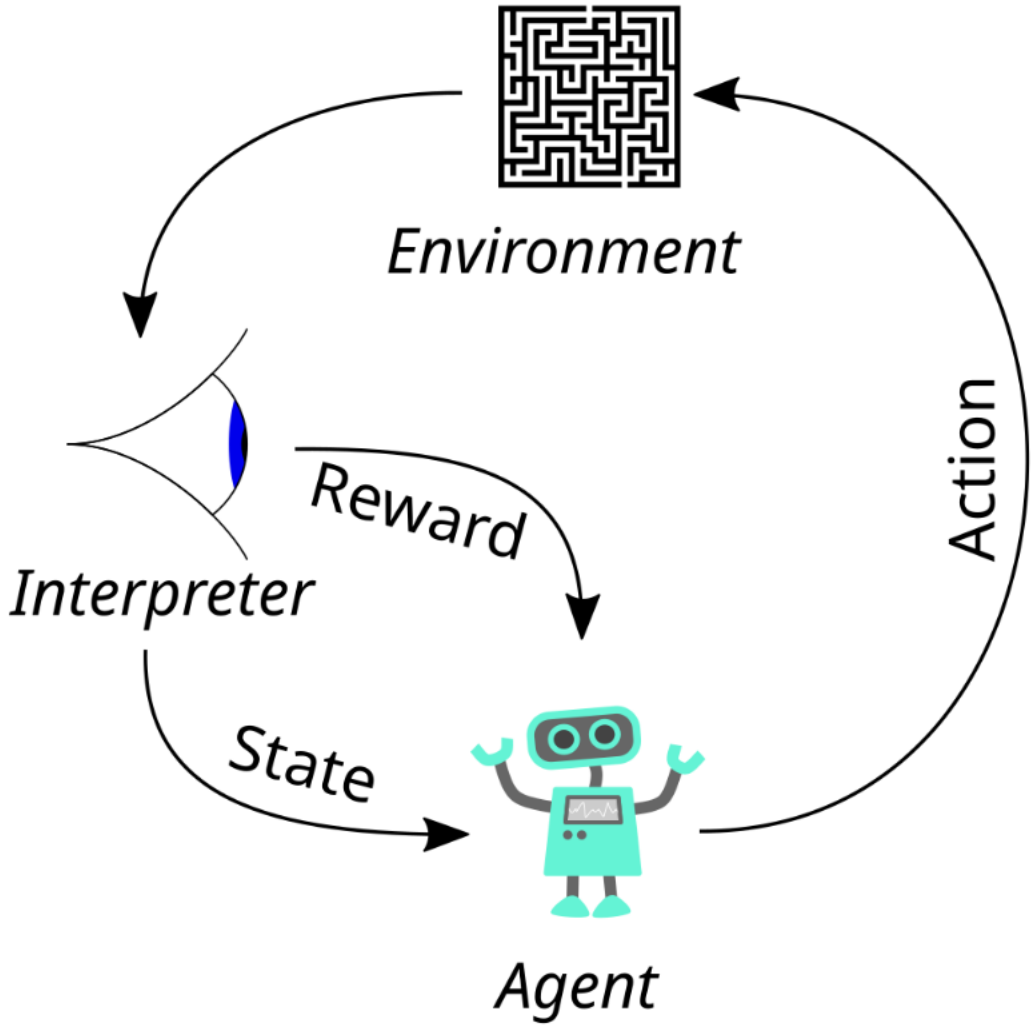


Class = f (input_pixel)



Reinforcement Learning

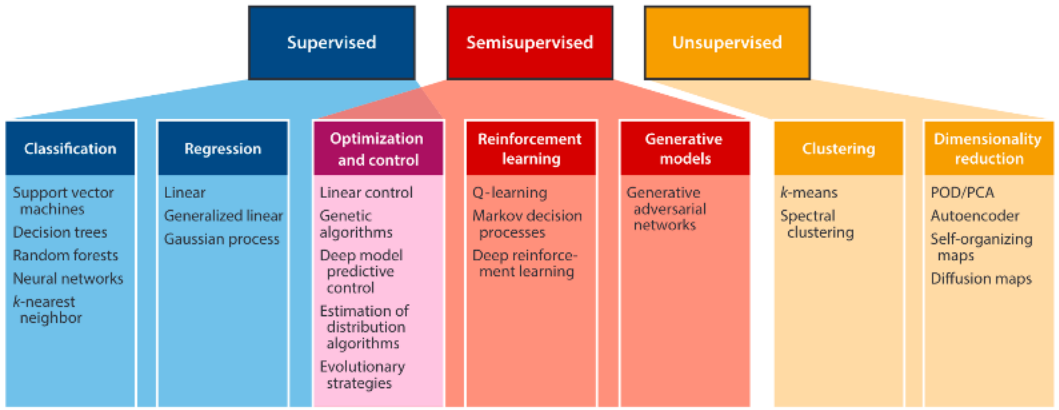




Examples:

- Maybe how humans learn
- Teach a computer to play a game

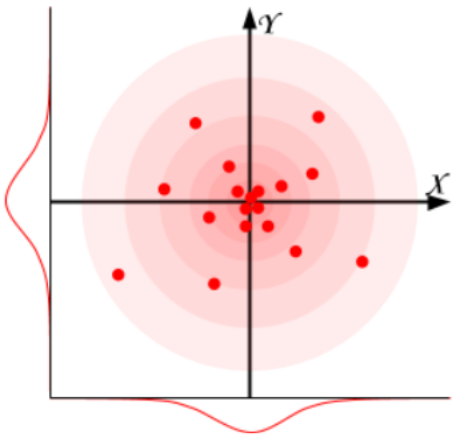
Generative Models



A generative model tries to learn the distribution from the limited data set, so we can then produce new samples that are similar to the original data.

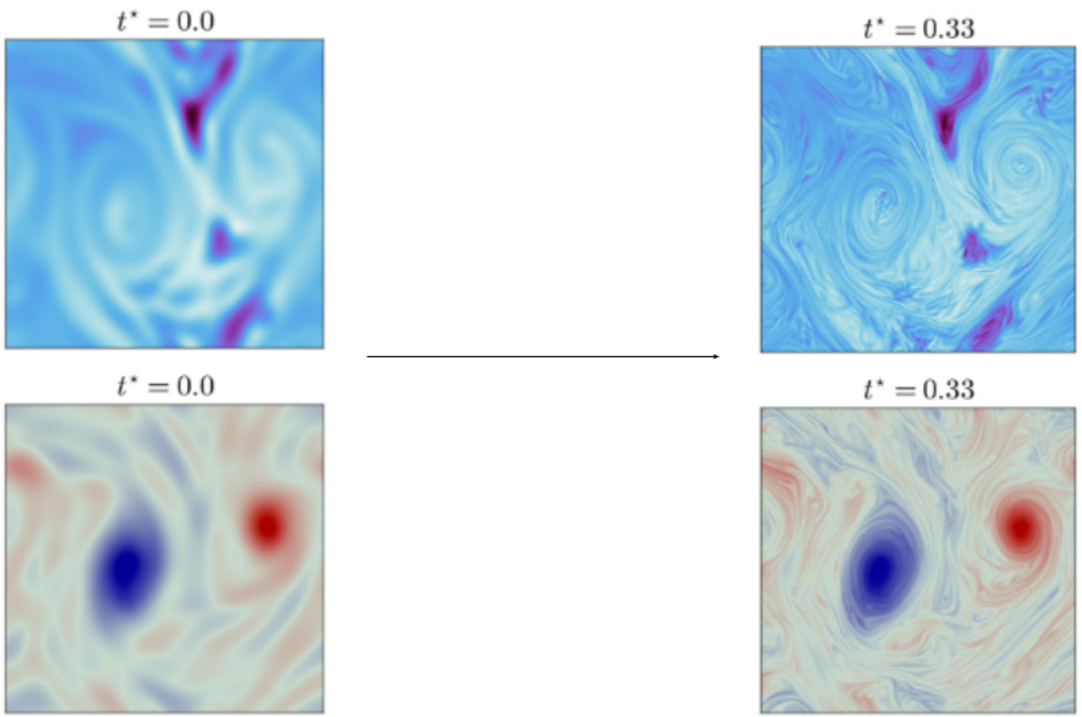
Climate relevant example:
- If we can only run a few high resolution simulations for CMIP, we could envision generating more “high resolution simulation like” outputs for estimating some statistics.

Knowing the distribution, I can generate samples.

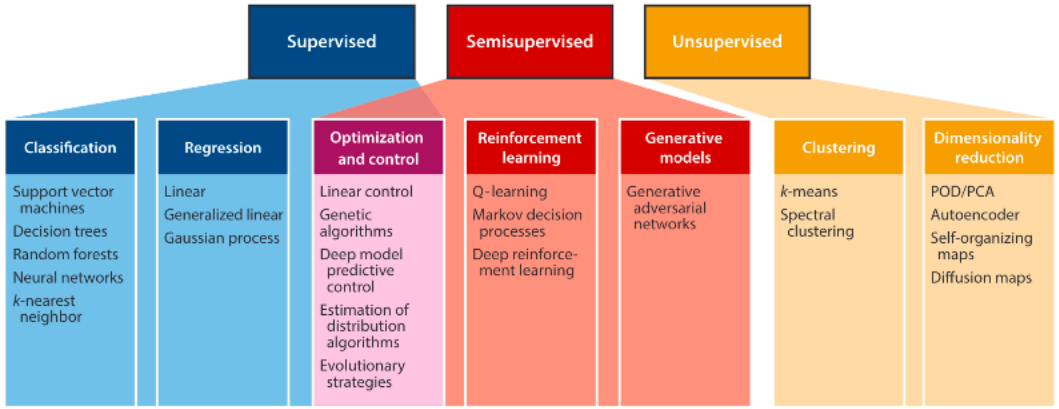


$$X \sim \mathcal{N}(0,1)$$

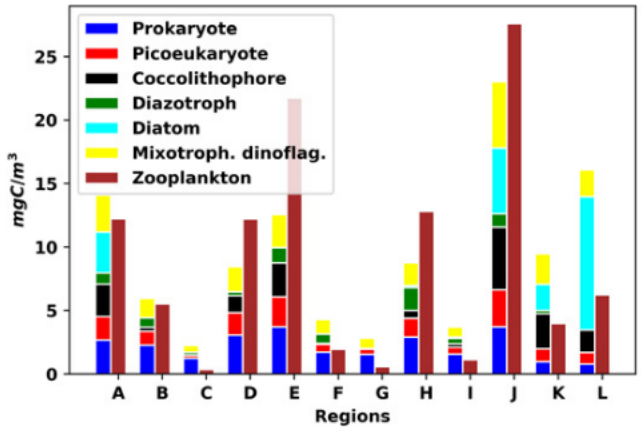
$$Y \sim \mathcal{N}(0,1)$$



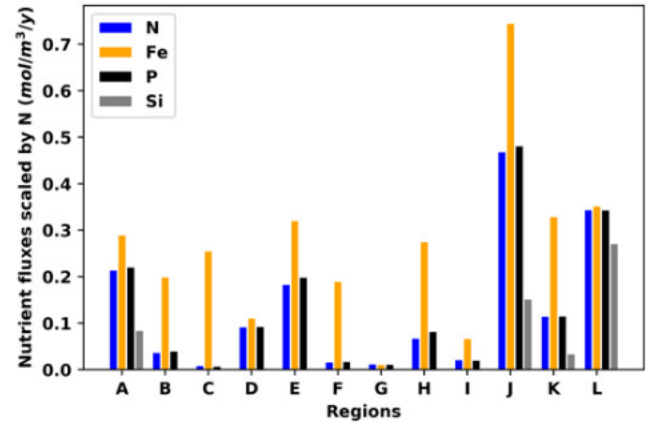
Clustering



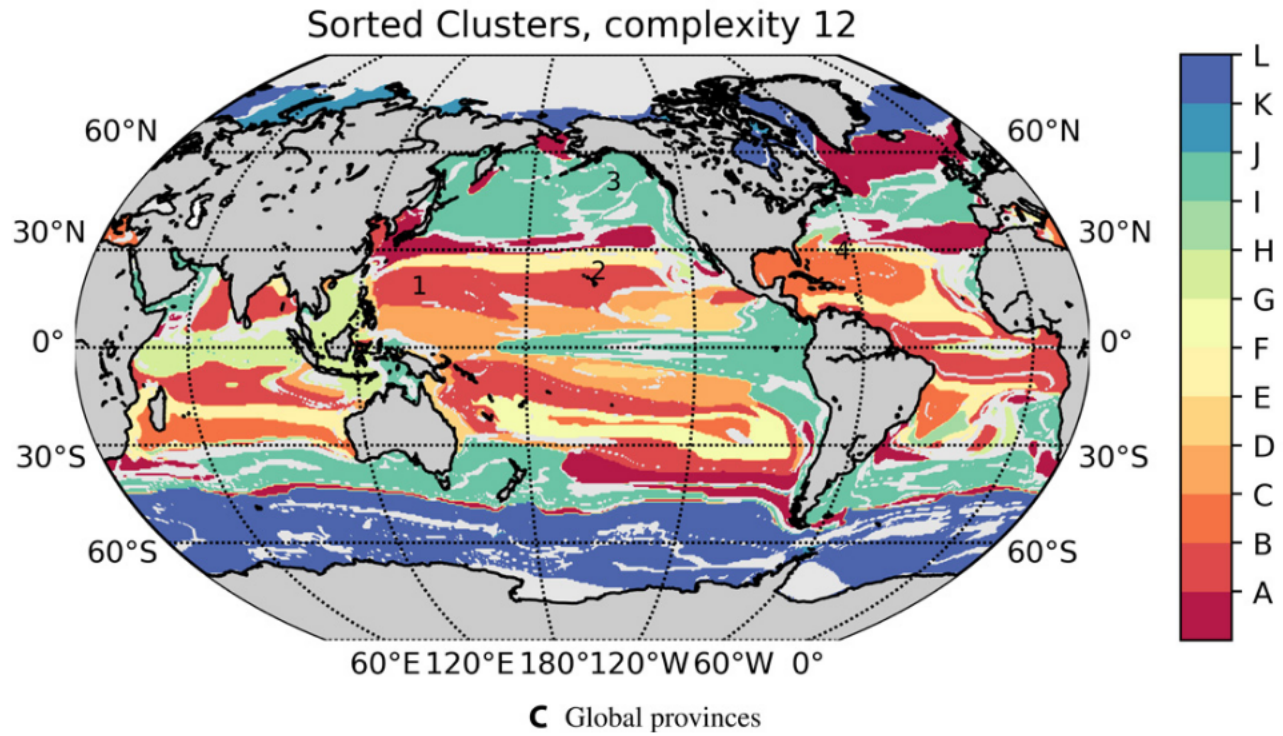
Automatically detect patterns in your data



A Ecological ensemble



B Nutrient fluxes

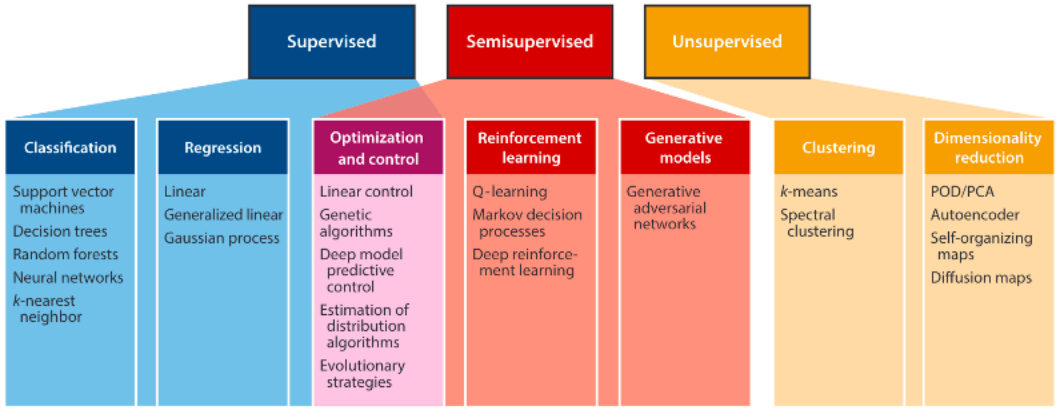


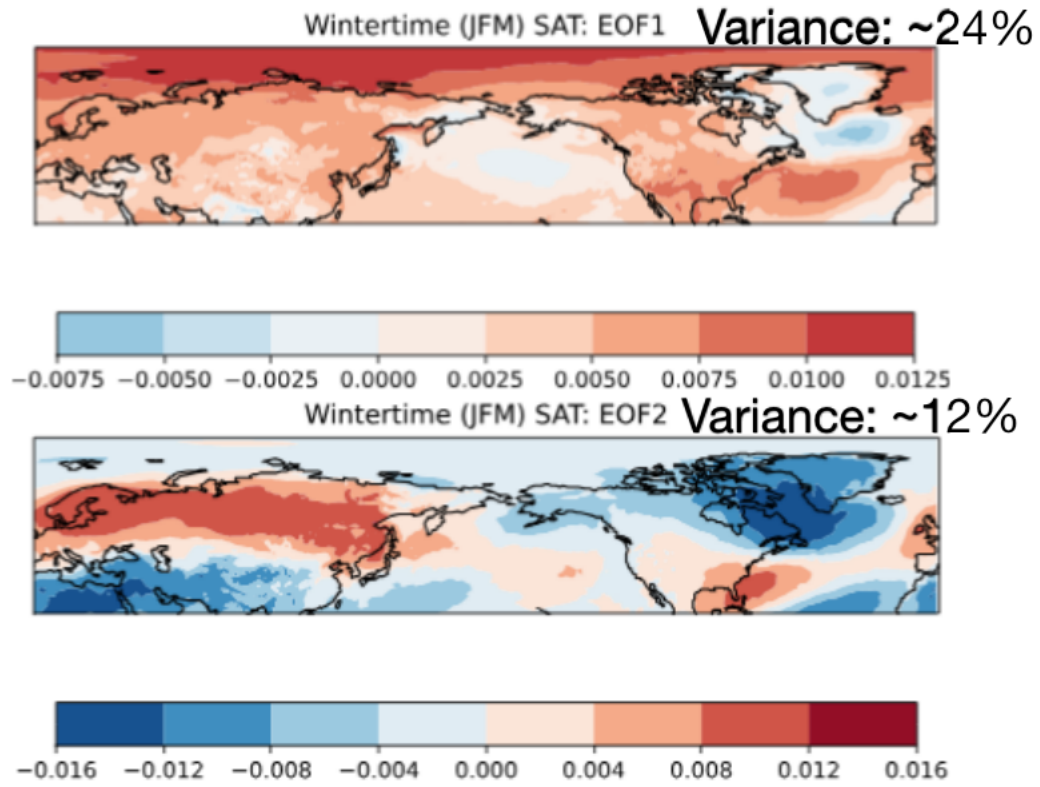
C Global provinces

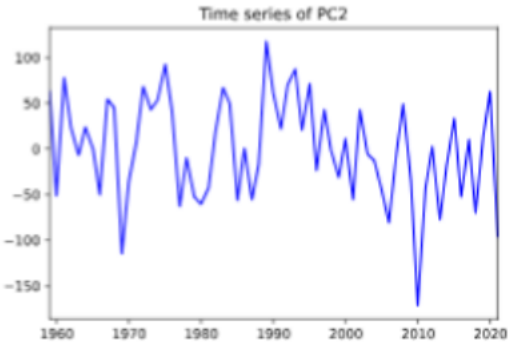
Elucidating ecological complexity: Unsupervised learning determines global marine eco-provinces

MAIKE SONNEWALD, STEPHANIE DUTKIEWICZ, CHRISTOPHER HILL, AND GAEL FORGET Authors Info & Affiliations

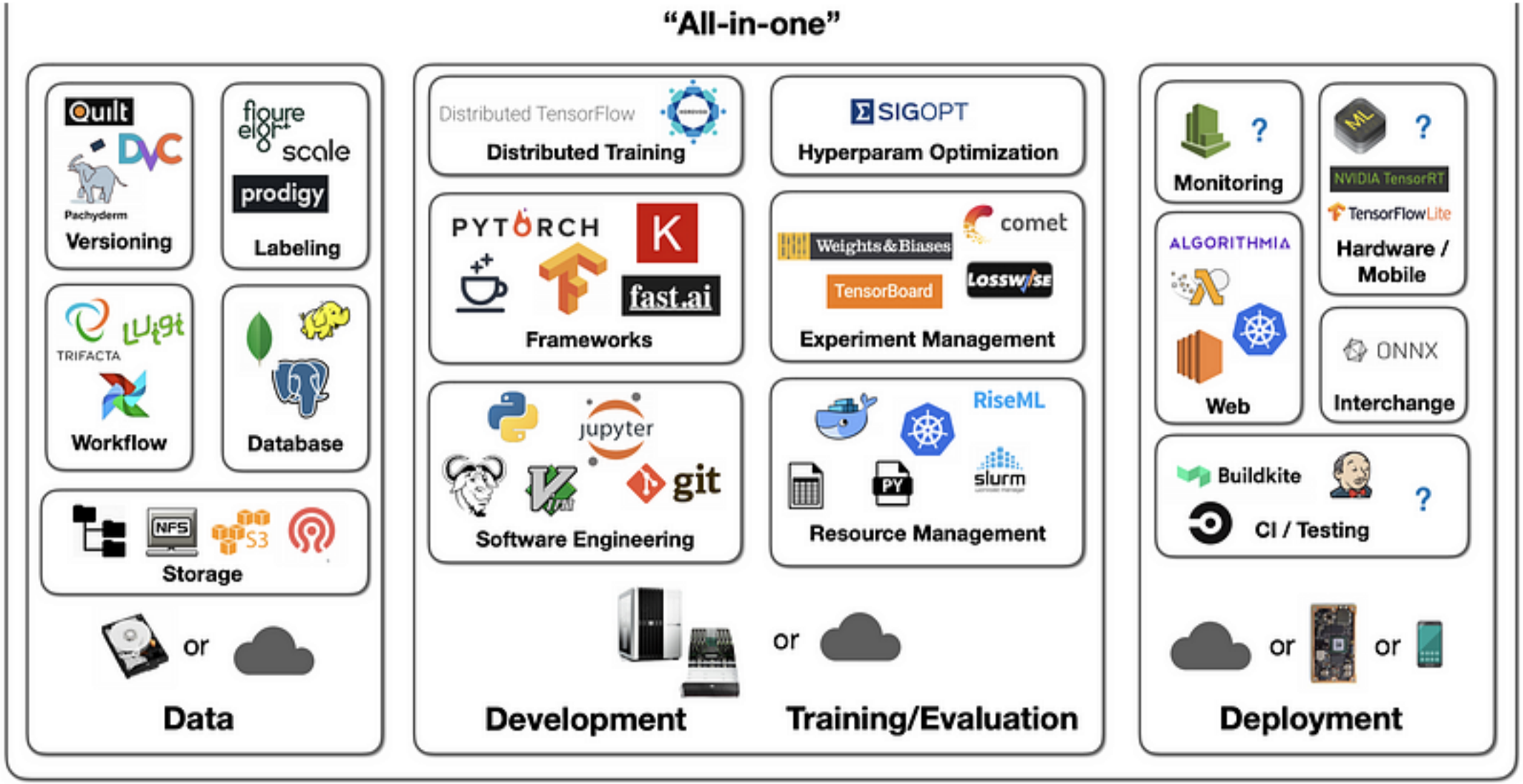
Dimensionality reduction







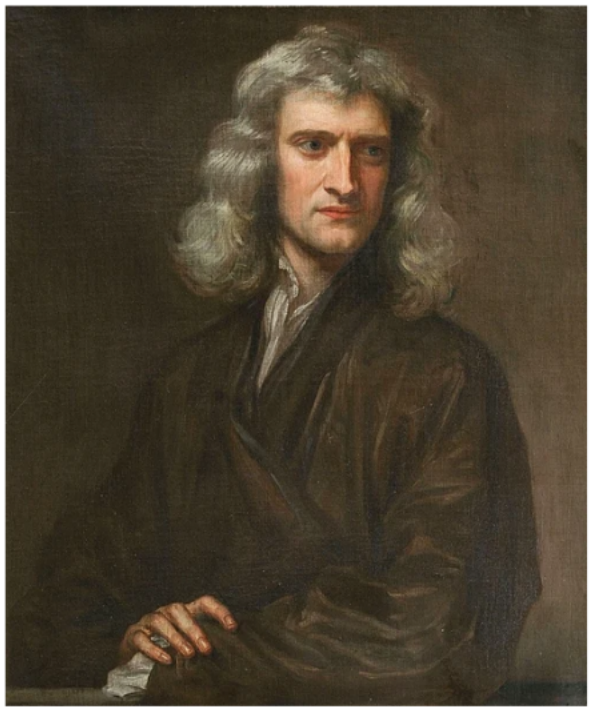
Vast range of software tools



All models are wrong
but some are useful

George E.P. Box

Function approximation from data (human learning)



Isaac Newton (1643-1727)

Newton's law of gravitation:

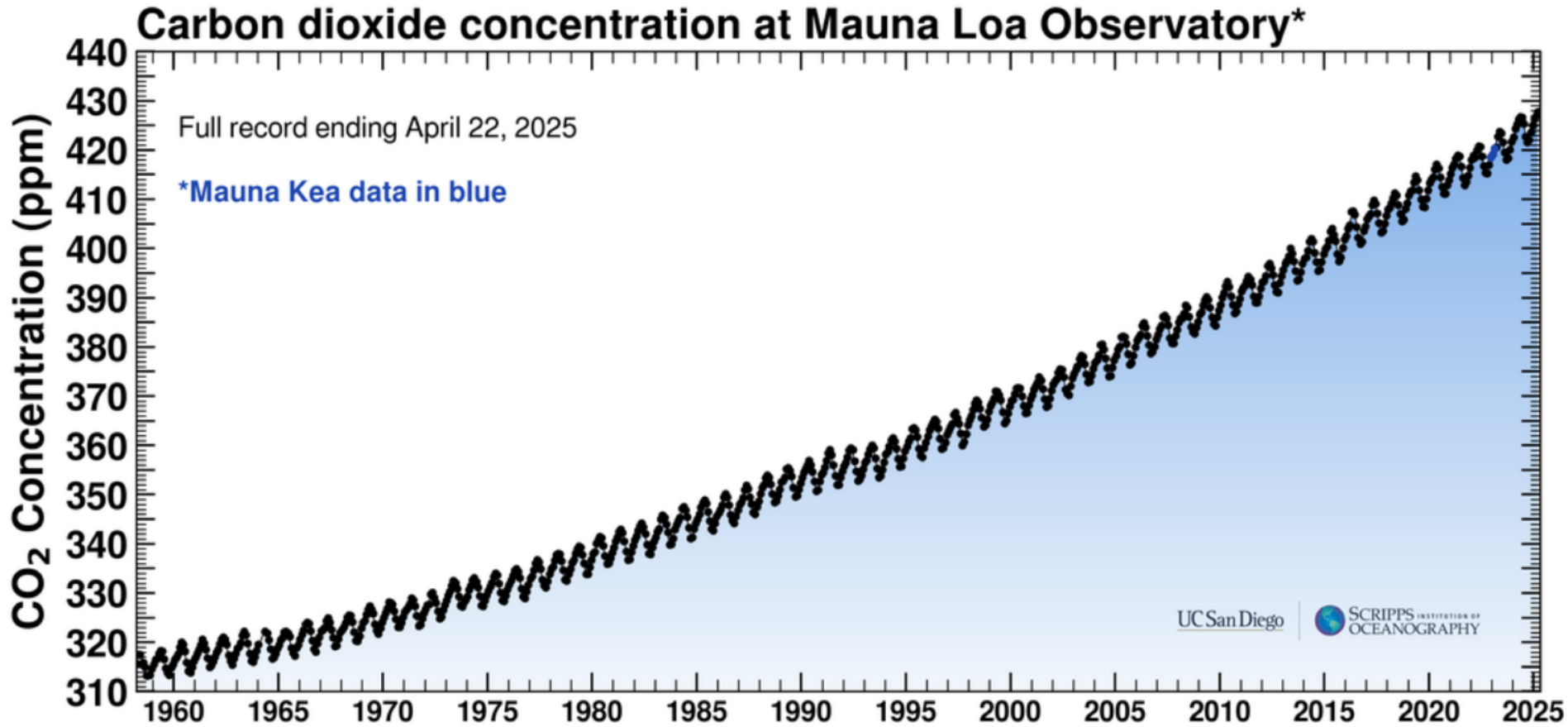
$$F = G\frac{m_1 m_2}{r^2} = f(m_1, m_2, r|G)$$

Function. 3 inputs. 1 parameter.

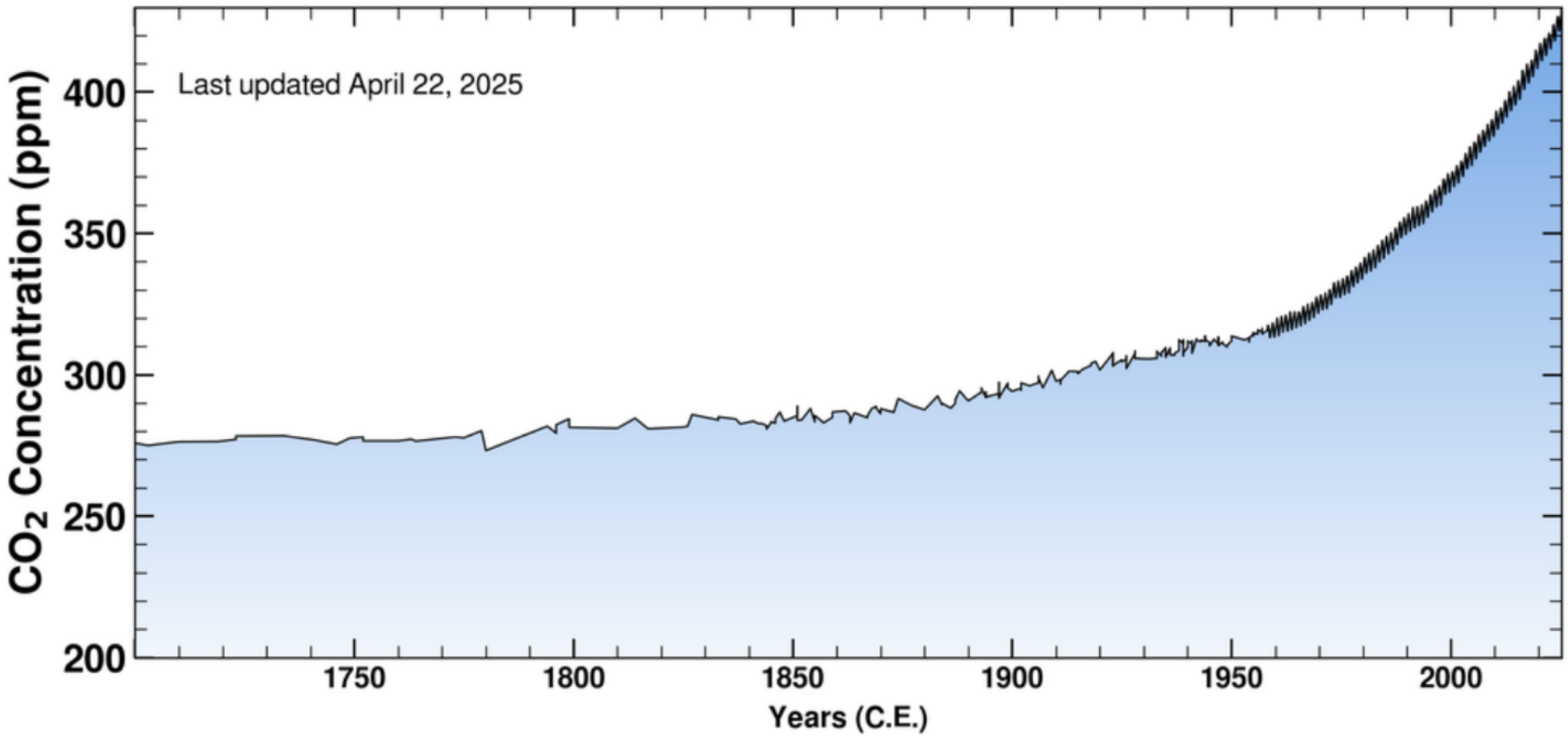
Successes:
- Single parameter
- Very wide range of applicability (apples to the planets)
- Generalizable:
  - Tycho Brahe's data to modern satellites

Failings (?):
- Precision at large scales, very fast speeds, very large objects
  - -> General relativity
- Very small scales
  - -> Quantum gravity (???)

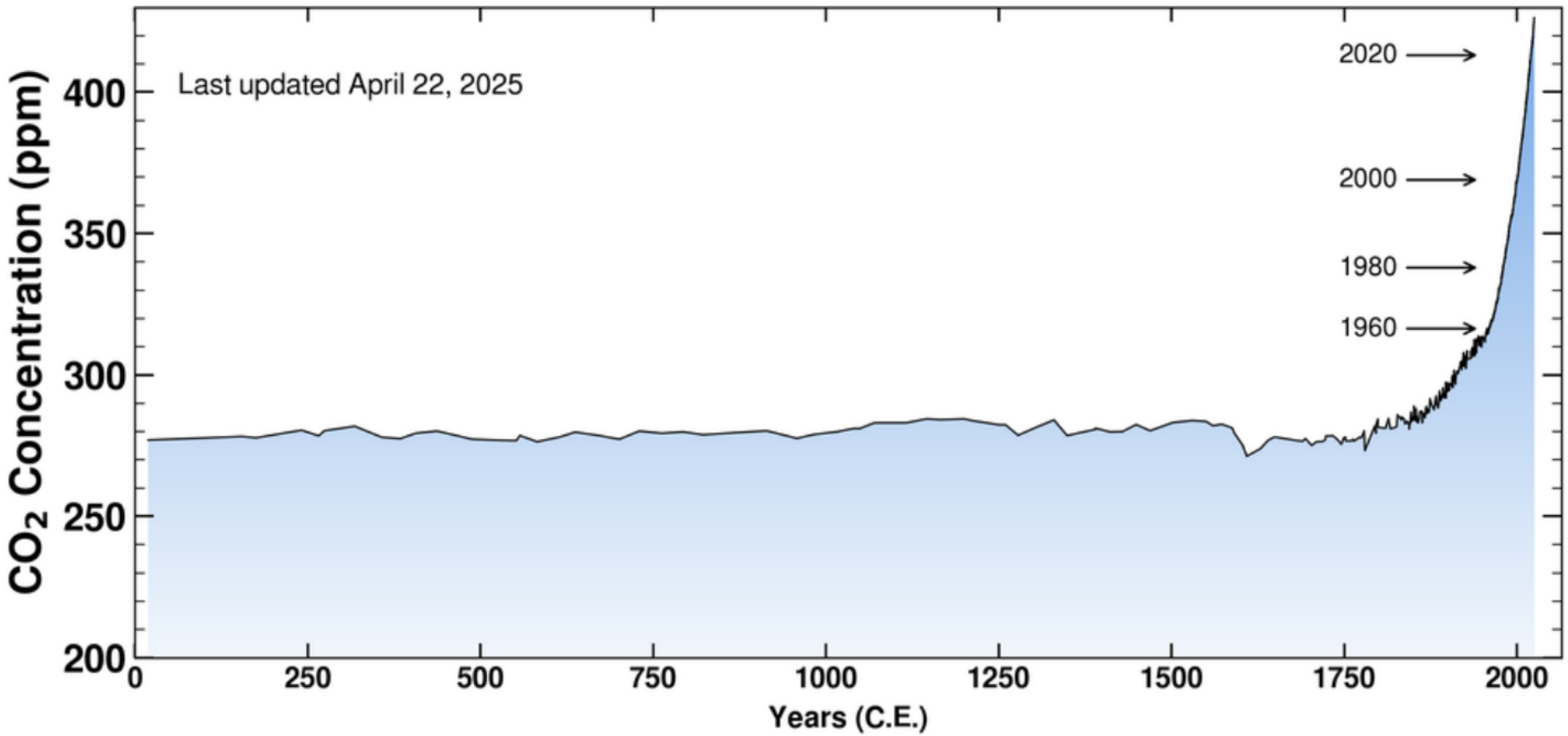
How much should we trust a model?



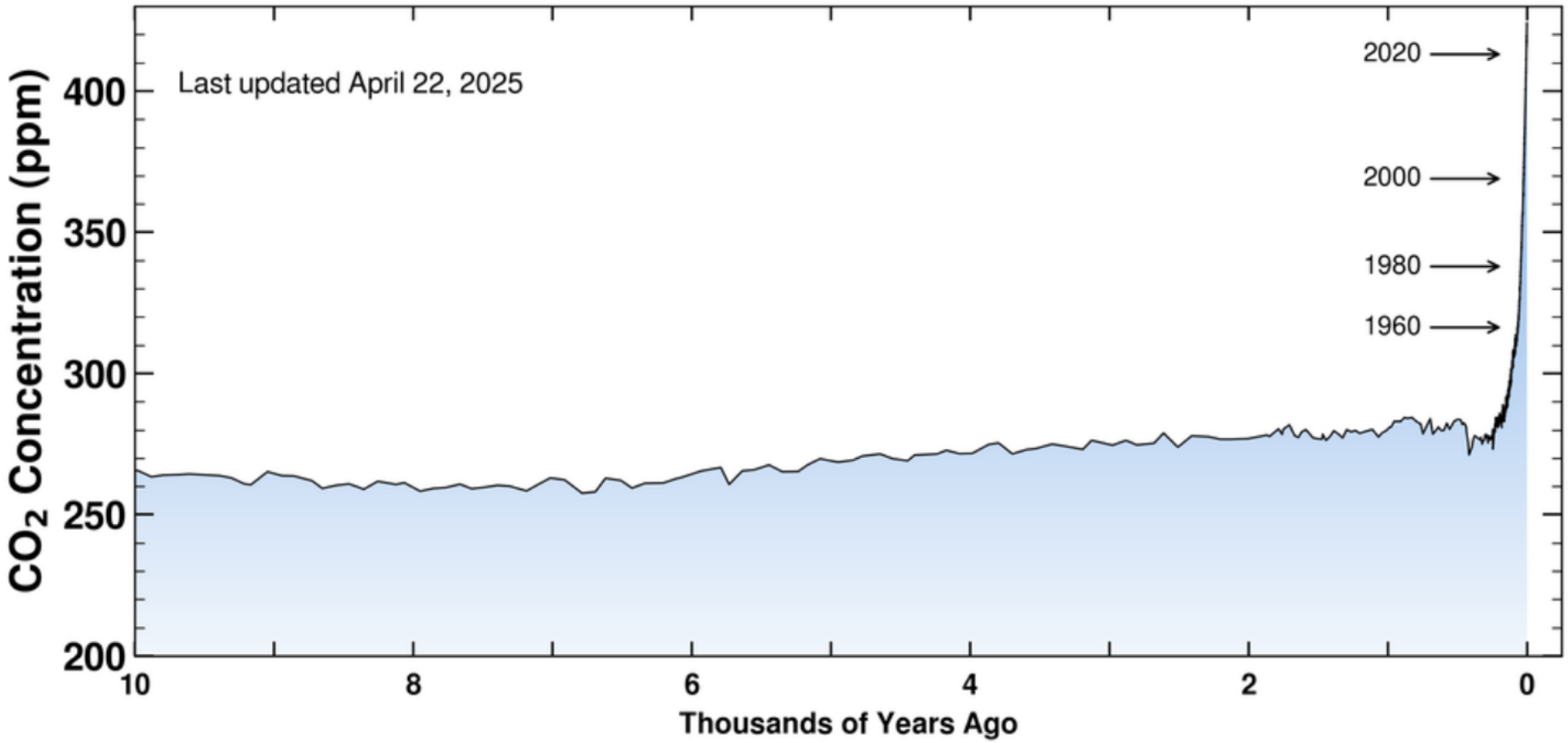
How much should we trust these models?



How much should we trust these models?



How much should we trust these models?



How much should we trust these models?